

Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

*Entwurf und Implementierung eines Echtzeit-Impedanzreglers für einen
7-achsigen kollaborativen Roboter*

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Declaration of Authorship

I hereby declare that this master's thesis entitled

*“Design and Implementation of a Real-Time Impedance Controller for a 7-DOF
Collaborative Robot”*

was written independently and without assistance from third parties. All sources, references, software tools, and other resources used have been cited accordingly. This work has not previously been submitted as an examination requirement in this or any other form.

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Munich, 15.09.2026

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Abstract

This thesis develops a real-time controller for a seven-degree-of-freedom collaborative robot intended for surface grinding with a rectangular tool on a planar surface. Cartesian impedance control is used to generate the required pressing motion while allowing passive, contact-induced rotation of the tool towards surface alignment. This behaviour is intended to reduce the effect of angular mismatch at contact entry, which can otherwise produce edge-first contact and increased contact loads.

Directional translational and rotational compliance was defined relative to the surface. In addition, a virtual centre of compliance could be displaced from the tool centre point to couple translational pressing with rotational motion and thereby influence the preferred direction of alignment.

Surface-contact experiments investigated the angular-error component about the first surface tangent relative to the calibrated surface. Different stiffness and centre-of-compliance settings were compared. Increasing rotational stiffness increased the angular error by about 276 %, whereas the tested tangential translational stiffness variation changed it by about 4 % of the baseline error. The effect of centre-of-compliance displacement depended on the entry direction. The lowest measured mean error magnitudes were about 53 % and 34 % below their respective tool-centre-point references, at different tested positions. The evaluated angular-error component could pass through zero. Further contact-induced rotation could increase its final magnitude.

A separate pose-hold experiment evaluated the null-space controller. Projected damping reduced cumulative joint motion by about 25 % relative to the inactive setting. Adding damping to singular-value conditioning reduced cumulative joint motion by about 51 % relative to conditioning alone.

Kurzfassung

Diese Arbeit entwickelt einen Echtzeitregler für einen siebenachsigen kollaborativen Roboter zur Oberflächenbearbeitung mit einem rechteckigen Werkzeug auf einer ebenen Fläche. Die kartesische Impedanzregelung erzeugt die Anpressbewegung und ermöglicht eine passive, kontaktinduzierte Drehung des Werkzeugs zur Ausrichtung an der Fläche. Dadurch sollen die Auswirkungen einer Winkelfehlstellung bei Kontaktbeginn reduziert werden, die andernfalls zu Kantenkontakt und erhöhten Kontaktlasten führen kann.

Die translatorische und rotatorische Nachgiebigkeit wurde richtungsabhängig zur Fläche definiert. Ein verschiebbares virtuelles Nachgiebigkeitszentrum koppelt Anpressbewegung und Rotation. Seine Position relativ zum Werkzeugbezugspunkt beeinflusst die bevorzugte Ausrichtungsrichtung.

In den Kontaktversuchen wurde die verbleibende Winkelfehlerkomponente um die erste Flächentangente relativ zur kalibrierten Flächenreferenz untersucht. Eine Erhöhung der Rotationssteifigkeit vergrößerte den verbleibenden Fehler um etwa 276 %. Die Variation der tangentialen Translationssteifigkeit veränderte ihn um etwa 4 % des Fehlers im Referenzfall. Die Wirkung der Verschiebung hing von der Richtung des Fehlers bei Kontaktbeginn ab. Die kleinsten gemessenen mittleren Fehlerbeträge lagen bei unterschiedlichen untersuchten Positionen etwa 53 % und 34 % unter den jeweiligen Referenzwerten mit dem Nachgiebigkeitszentrum am Werkzeugbezugspunkt. Die Winkelfehlerkomponente konnte ihr Vorzeichen wechseln. Weitere kontaktinduzierte Drehung konnte ihren verbleibenden Betrag vergrößern.

In einem getrennten Posehalteversuch wurde die Nullraumregelung untersucht. Die projizierte Dämpfung verringerte die kumulierte Gelenkbewegung gegenüber der Einstellung ohne Nullraumdrehmoment um etwa 25 %. Die Kombination aus Dämpfung und Singulärwertkonditionierung verringerte die kumulierte Gelenkbewegung gegenüber der alleinigen Konditionierung um etwa 51 %.

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List of Abbreviations

CoC	Centre of Compliance
CSV	Comma-Separated Values
DH	Denavit–Hartenberg
DOF	Degree of Freedom
EE	End-Effector
E-stop	Emergency Stop
FCI	Franka Control Interface
FCU	Franka Control Unit
F/T	Force/Torque
IP	Internet Protocol
LDLT	Lower–Diagonal–Lower-Transpose Factorisation
OS	Operating System
PREEMPT_RT	Real-Time Preemption Patch
ROS	Robot Operating System
SE(3)	Special Euclidean Group in three dimensions
SO(3)	Special Orthogonal Group in three dimensions
SVD	Singular Value Decomposition
TCP	Tool Centre Point

List of Symbols

Cartesian quantities are expressed in the robot base frame $\{0\}$ unless a frame suffix says otherwise, and the base-frame index is omitted. Suffixes $,S$ and $,EE$ indicate the surface and end-effector frames.

Frames and Kinematics

$\{0\}$	Robot base frame	$[-]$
$\{EE\}$	End-effector frame used by the robot model	$[-]$
$\{d\}$	Desired end-effector and TCP reference frame	$[-]$
$\{S\}$	Surface frame, with axes t_1 , t_2 and n_s	$[-]$
$q, \dot{q}$	Joint position / velocity vector	$[\text{rad}], [\text{rad/s}]$
q_{init}	Configured initial joint posture	$[\text{rad}]$
p	Position / translation vector	$[\text{m}]$
R	Rotation matrix	$[-]$
T	Homogeneous transformation matrix	Rotation block: $[-]$; translation block: $[\text{m}]$
$J(q)$	Geometric Jacobian; linear rows above angular rows	$[\text{m/rad}], [-]$

Pose and Error

p_{EE}, p_{TCP}	End-effector / tool-centre-point position; $p_{TCP} = p_{EE}$ in the implemented model	[m]
R_{EE}, T_{EE}	End-effector orientation / transformation	R : [-]; T : rotation block [-], translation block [m]
$R_{EE,clearance}, p_{TCP}(t_{start})$	End-effector orientation captured at the clearance transition and held as the contact-establishment orientation reference / TCP position captured at the same instant	[-], [m]
p_d, R_d, T_d	Desired position / orientation / transformation	p : [m]; R : [-]; T : rotation block [-], translation block [m]
ΔR	Current-to-desired relative rotation in the base frame, $R_d R_{EE}^\top$	[-]
ϕ	Orientation-error angle	[rad]
$u, [u]_\times$	Base-frame orientation-error axis / its skew matrix	[-]
e_p	Translational (position) error	[m]
e_R	Base-frame rotational error used by the command	[rad]
$\dot{p}_d, \omega_d$	Desired end-effector linear / angular velocity; $\omega_d = 0$ in the implemented controller	[m/s], [rad/s]
$\dot{p}_{EE}, \omega_{EE}$	Measured end-effector linear / angular velocity	[m/s], [rad/s]

Cartesian Stiffness and Damping

K_p, D_p	Generic translational stiffness / damping block; a frame or state index is added when needed	$[\text{N/m}], [\text{N s/m}]$
K_R, D_R	Generic rotational stiffness / damping block; a frame or state index is added when needed	$[\text{N m/rad}], [\text{N m s/rad}]$
$K_{p,t_1}, K_{p,t_2}, K_{p,n}$	Translational stiffness entries along t_1, t_2 , and n_s	$[\text{N/m}]$
$K_{R,t_1}, K_{R,t_2}, K_{R,n}$	Rotational stiffness entries about t_1, t_2 , and n_s	$[\text{N m/rad}]$
$D_{p,t_1}, D_{p,t_2}, D_{p,n}$	Translational damping entries along t_1, t_2 , and n_s	$[\text{N s/m}]$
$D_{R,t_1}, D_{R,t_2}, D_{R,n}$	Rotational damping entries about t_1, t_2 , and n_s	$[\text{N m s/rad}]$
K, D	Complete base-frame Cartesian stiffness / damping, assembled block diagonal from the rotated translational and rotational blocks	$[\text{N/m}], [\text{N m/rad}];$ $[\text{N s/m}], [\text{N m s/rad}]$

Dynamics and Torques

$M(q)$	Joint-space inertia matrix	[kg m ²]
$\Lambda(q), \Lambda_S(q)$	Regularised operational-space inertia in the base frame / resolved along the surface-frame directions	Translational: [kg]; coupling: [kg m]; rotational: [kg m ²]
$\lambda_i(q)$	Directional Cartesian inertia, the i th diagonal entry of $\Lambda_S(q)$	Translational: [kg]; rotational: [kg m ²]
k_i, d_i	Directional stiffness / inertia-scaled damping coefficient	Translational: [N/m], [N s/m]; rotational: [N m/rad], [N m s/rad]
ζ	Selected damping factor	[-]
F	Complete commanded Cartesian wrench	[N], [N m]
f, m	Complete commanded Cartesian force / moment at the TCP	[N], [N m]
F_n, f_n, f_t	Commanded normal-force component, $F_n = n_s^\top f$ / normal force vector, $f_n = F_n n_s$ / tangential force vector, $f_t = f - f_n$	[N]
M_{t_1}	Commanded TCP moment component about t_1 , $M_{t_1} = t_1^\top m$	[N m]
$F_{n,\text{est}}, M_{t_1,\text{est}}$	Model-estimated external normal-force component, $F_{n,\text{est}} = n_s^\top f_{\text{est}}$ / model-estimated external TCP moment about t_1 , $M_{t_1,\text{est}} = t_1^\top m_{\text{est,TCP}}$	[N], [N m]
τ	Joint torque vector	[N m]
τ_c	Coriolis and centrifugal compensation torque	[N m]

τ_{cart}	Cartesian impedance torque, $J^\top(q)F$	[N m]
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τ_{cmd}	Complete commanded joint torque	[N m]
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Surface and Contact Geometry

n_s	Configured surface-reference normal copied into the controller and used in the evaluation	[-]
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n_{phys}	Physical surface normal	[-]
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t_1, t_2	Surface tangent directions (unit)	[-]
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R_{surface}	Surface-frame orientation $[t_1 \ t_2 \ n_s]$ in the base frame	[-]
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$n_{\text{Tool,EE}}, n_{\text{Tool}}, n_{\text{Tool},d}$	Tool-face normal expressed in the EE frame, obtained from calibration / the same direction expressed in the base frame, $n_{\text{Tool}} = R_{\text{EE}}n_{\text{Tool,EE}}$ / desired tool-normal direction used by the controller	[-]
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$r_{\text{face,EE}}$	Calibrated tool-face-centre offset from the TCP in end-effector coordinates	[m]
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θ_{offset}	Configured pre-contact orientation offset that sets the desired tool direction, formed to first order from its two tangent contributions	[rad]
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p_s	Point on the configured surface	[m]
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e_{x_0}, e_{z_0}	Unit vectors along the base-frame axes x_0 and z_0	[-]
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p_c	Centre of compliance	[m]
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r_c	TCP-to-compliance-centre displacement, $p_c - p_{\text{TCP}}$, as the point-shift transformation and the experimental plots and tables use it	[m]
$r_{c,S}, r_{c,\text{EE}}$	The same displacement represented in the surface and end-effector frames, with $r_c = R_{\text{surface}} r_{c,S} = R_{\text{EE}} r_{c,\text{EE}}$	[m]
$r_{c,t_1}, r_{c,t_2}, r_{c,n}$	Surface-frame components of r_c along t_1, t_2 and n_s	[m]
$r_{\text{Tool,EE}}, p_{\text{Tool}}$	Selected tool-point offset from the TCP, expressed in the EE frame / selected tool-point position in the base frame	[m]
$p_{\text{Tool}}(t_{\text{start}}), p_{\text{Tool},0}, p_{\text{Tool},d}$	Selected tool point at the start of Contact Establishment / its projection onto the configured surface reference / its desired contact-establishment position	[m]
$s_a, v_a, s_{a,\text{max}}$	Commanded approach travel of the TCP along $-n_s$ / its configured rate / its configured maximum	[m], [m/s], [m]
$p_{\text{TCP,start}}$	TCP position captured when surface approach begins	[m]
$t_{\text{start}}, t_{\text{end}}$	Instants at which the state under discussion begins and ends; used for Surface Approach and for Contact Establishment	[s]
p_i	Base-frame position of tool-face corner i	[m]

h_i	Surface-normal clearance of tool-face corner i from the configured surface reference	[m]
h_{Tool}	Minimum clearance of the rectangular tool face from the configured surface reference, $h_{\text{Tool}} = \min_i h_i$	[m]
$h_{\text{clearance}}$	Configured tool clearance at which the transition from approach to contact establishment occurs	[m]
ε_{sel}	Tool-point selection tolerance	[m]
I_{sel}, N	Set of tool-face corners within the selection tolerance of the minimum clearance / its size	[-]
$\theta_{\text{app,err}}, \varepsilon_{\text{app}}$	Tool-orientation angular error during Tool Orientation / its configured tolerance	[rad]
$s_c, v_c, s_{c,\text{max}}$	Commanded contact-establishment press coordinate of the selected tool point along $-n_s$ / its configured rate / its configured endpoint	[m], [m/s], [m]
m_R	Rotational-impedance contribution to the commanded moment, $K_R e_R - D_R \omega_{\text{EE}}$, present at every compliance-centre position	[N m]

Compliance and Alignment

$\text{Ad}(r_c)$	TCP-to-compliance-centre point-shift adjoint	Diagonal blocks: [-]; point-shift block: [m/rad]
K_c, D_c	Block-diagonal stiffness / damping at the centre of compliance	K : [N/m], [N m/rad]; D : [N s/m], [N m s/rad]
$K_{\text{TCP}}, D_{\text{TCP}}$	Centre-of-compliance stiffness and damping mapped to the TCP by $\text{Ad}(r_c)$	K : [N/m], [N/rad], [N m/m], [N m/rad]; D : [N s/m], [N s/rad], [N m s/m], [N m s/rad]
$\theta_{t_1}, \theta_{t_2}, u$	Rotations of a general angular tool offset about t_1 and t_2 in the compliance-centre direction rule / the unit direction of their first-order combination in the surface tangent plane	[rad], [rad], [-]
$\theta_{\text{offset}, t_1}, \theta_{\text{offset}, t_2}$	First-order contributions of the configured rotations about t_1 and t_2	[rad]
θ_{meas}	Pose-based rotation vector from the inward configured surface normal to the calibrated tool normal at contact entry	[rad]
$\theta_{\text{meas}, t_1}$	Measured pose-based angular offset about t_1 , relative to the configured reference	[rad]
$\theta_{\text{err}}(t)$	Pose-based angular-error vector from the inward calibrated surface normal to the tool normal calculated from measured end-effector orientation	[rad]

$\theta_{\text{err},t_1}(t)$	Angular-error component about t_1 relative to the calibrated surface reference	[rad]
$r_{c,t}$	Tangential part of the compliance-centre displacement, $r_{c,t_1}t_1 + r_{c,t_2}t_2$; its magnitude is written $\ r_{c,t}\ $	[m]
Null-Space and Singular Value Decomposition		
J^+	Thresholded-SVD Moore–Penrose pseudoinverse	[rad/m], [-]
N_q, N_τ	Joint-velocity / torque null-space projector	[-]
$\sigma_i, \sigma_{\min}$	Jacobian singular value / σ_6 conditioning indicator. The geometric Jacobian mixes linear and angular rows, so a singular value carries no single physical unit and its numerical value depends on the chosen Jacobian representation and length unit; it is used as a relative indicator under that convention	Representation- dependent
v_7	Structural null direction of the seven-joint Jacobian	[-]
$\dot{q}_{\text{null}}$	Projected joint velocity in the numerical null space	[rad/s]
d_{null}	Null-space damping	[N m s/rad]
k_σ	Sigma-conditioning torque magnitude	[N m]

α_{probe}	Null-direction sampling angle	[rad]
q_+, q_-	Joint configurations sampled either side of the current one along the null direction	[rad]
$\sigma_+, \sigma_-, \Delta_\sigma$	Minimum singular values at the two sampled configurations / their difference $\sigma_+ - \sigma_-$	Representation-dependent
s_σ	Conditioning direction selector, +1, -1 or 0 from Δ_σ against the deadband	[-]
ρ, δ_σ	Relative SVD cutoff / sampled-singular-value comparison deadband	[-], Representation-dependent
$\tau_d, \tau_\sigma, \tau_{\text{null}}$	Projected damping / sigma / total null-space torque	[N m]
E_N	Cumulative projected joint motion across all seven joints	[rad]
Δq_1	Measured joint-1 angle change from disturbance onset	[rad]
$u_f, F_{\text{dist}}, s_{\text{dist}}$	Virtual-disturbance direction / force magnitude / waveform	[-], [N], [-]
$f_{\text{dist}}, \tau_{\text{dist}}$	Virtual-disturbance force / equivalent joint torque, $J^\top f_{\text{dist}}$	[N], [N m]
$\ e_p\ _{\text{max}}$	Maximum Cartesian position error over the disturbance interval	[m]

Chapter 1

Introduction

1.1 Motivation

In surface-contact tasks, the commanded tool pose is defined relative to a configured surface reference with normal n_s . Placement and calibration tolerances can make the physical surface normal n_{phys} differ from that configured reference. The tool orientation achieved before contact can also differ from the desired orientation relative to the same reference.

Together, the difference between the configured and physical surfaces and the difference between the desired and achieved tool orientations determine the angular mismatch between the tool face and the physical surface at contact entry. That mismatch can bring the tool onto the surface along an edge or at a corner. A rigid position controller opposes the corresponding corrective rotation and can therefore generate high contact loads.

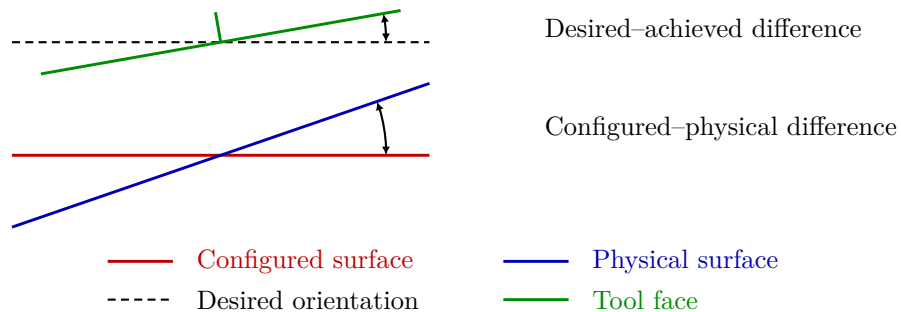


Figure 1.1: Sources of angular offset at surface entry.

Figure 1.1 separates the two contributions to that mismatch: the difference between the configured and physical surfaces, and the difference between the desired and achieved tool

orientations.

Torque-controlled collaborative robots provide the interface required for active Cartesian compliance [11, 1]. Joint torque sensing and a torque-level command interface allow the controller to regulate both Cartesian motion and compliance during contact. In surface finishing, this compliance can limit the loads caused by angular tool–surface mismatch. It can also permit end-effector rotation towards parallel alignment with the configured surface under the commanded normal press. The angular error during Contact Establishment is evaluated between the calibrated surface reference and the tool normal calculated from measured end-effector orientation.

Impedance control defines this compliant behaviour as a dynamic relation between motion and interaction wrench [11]. Cartesian impedance control applies that relation to the end-effector pose. Translational and rotational stiffness and damping can be assigned independently along selected Cartesian directions. During edge-first contact, the wrench contains both force and moment components. Finite rotational compliance allows the contact moment to rotate the end effector, while the translational impedance generates the commanded press. The selected stiffness and damping matrices therefore determine both the angular behaviour during contact and the free-space pose-tracking behaviour.

1.2 Related Work

Cartesian impedance control builds on standard serial-manipulator kinematics [19, 3, 16]: homogeneous transformations for representing the end-effector pose and the Jacobian for relating joint motion to Cartesian motion. Khatib established the operational-space formulation, in which a Cartesian wrench F is mapped to joint torques τ through $J^\top$ [12]; this mapping is adopted in the controller. Hogan defined the impedance relation between end-effector motion and interaction force [11]. Albu-Schäffer et al. implemented Cartesian impedance control using joint-torque sensing [1], which corresponds to the torque-controlled interface used in this thesis. These formulations supply the kinematic and control basis adopted here.

Mayr et al. implemented a modular Cartesian impedance controller for the Franka Emika Panda that combines real-time model access, null-space control, signal treatment, and

command limiting [14]. Their implementation identifies the real-time components required on the robot platform used here.

Recent work has applied compliant and force-controlled strategies directly to robotic surface finishing. Alt et al. combined perception, planning, and force-controlled execution for robotic surface treatment [2]. Min et al. integrated demonstrated motion and force skills with impedance control on a 7-DOF collaborative robot [15]. Wu et al. regulated the contact force during robotic grinding with adaptive variable impedance control [24]. In each of these studies the regulated quantity is the commanded contact force rather than the rotational response at contact entry.

Fasse and Broenink formulated spatial impedance about a selected point, so that translation and rotation become coupled through the position of this point [5]. This formulation provides the theoretical basis for the compliance-centre shift used in this thesis. The same principle appears mechanically in remote-centre-compliance devices. Their elastic elements are arranged so that the tool behaves as if it rotates about a virtual point located away from the mechanism itself [17, 23]. More recent mechanically compliant and variable-compliance-centre strategies apply this principle to peg-in-hole insertion [22, 10]. In those mechanisms, elastic deformation and mechanism geometry permit the contact forces to generate corrective motion that reduces misalignment. Most of these studies address insertion tasks and fix the compliance centre through either the mechanical design or a strategy adapted to the hole geometry.

Suomalainen et al. experimentally investigated how compliance-centre placement and compliant-axis selection influence alignment in an assembly task [21]. Friedrich et al. subsequently proposed an active remote centre of compliance (CoC) that combines passive compliance with active force control [9]. Both show that the location of the compliance centre can strongly influence the direction and the magnitude of contact-induced motion.

Together, the cited work provides the basis for compliant surface treatment, spatial impedance about a selected point, and compliance-centre placement for assembly. In this thesis, the effects of directional Cartesian stiffness and a software-defined virtual compliance centre were isolated during planar tool–surface contact.

1.3 Problem Statement

Before contact, the tool–surface angular relation depends on differences between the configured and physical surfaces and between the desired and achieved tool orientations. The intended application can encounter both directions of rotation about a surface-tangent axis and different rotation axes within the tangent plane. Under rigid position control, the contact geometry can generate high moments while the controller resists the corrective motion. Finite rotational compliance, however, allows contact-induced rotation, while a displaced CoC additionally couples translation and rotation.

The displacement r_c points from the tool centre point (TCP) to the virtual CoC. It contributes the term $r_c \times f$ to the commanded TCP moment. For the commanded normal press, the tangential direction of r_c sets the direction of that additional moment. A displaced centre therefore favours one rotational direction and opposes the other.

Different CoC positions were evaluated for angular offsets in both directions of rotation about one axis in the surface plane. The comparison determined how each centre position changed the measured end-effector response in the two rotational directions.

The selected controller parameters were varied while the calibrated geometry, robot configuration, and tool mounting were held constant. A configured orientation offset sets the desired tool direction relative to the configured surface reference. The orientation achieved at contact entry characterises the angular condition of each experiment and accounts for the finite orientation tolerance. Chapters 2 and 3 introduce the frame conventions and controller formulation used in the experiments.

1.4 Contributions

The main contributions are:

- A real-time Cartesian impedance controller with surface-relative directional stiffness and damping and a shiftable virtual CoC was implemented on a Franka Emika Panda.
- The effects of rotational stiffness, translational stiffness along the surface, and tangential compliance-centre position on contact-induced end-effector rotation were

characterised experimentally.

- Projected null-space damping and singular-value conditioning were implemented and evaluated individually and together in Cartesian pose hold.

With the CoC at the TCP, the measured end-effector rotation was towards the configured surface for both tested rotational directions. A displaced CoC favoured one rotational direction and opposed the other through the additional commanded moment $r_c \times f$. Selecting the displacement direction according to the angular-offset direction slightly increased the final response, whereas the opposite displacement direction strongly reduced it.

Chapter 2

Theoretical Background

This chapter introduces the mathematical formulation used by the Cartesian impedance controller. The frame convention is defined first because the following Cartesian quantities depend on their coordinate representation.

Unless stated otherwise, Cartesian vectors, orientations, velocities, wrenches, Jacobians, and stiffness and damping matrices are expressed in the robot base frame $\{0\}$, and the base-frame index is omitted. A quantity represented in another frame carries an explicit frame suffix, S for the surface frame and EE for the end-effector frame. The compliance-centre displacement r_c is therefore the base-frame representation, while $r_{c,S}$ and $r_{c,EE}$ are the same physical displacement in surface and end-effector coordinates.

A suffix that names a point or a body is not a frame suffix. In p_{EE} the index identifies the point whose position is given, and that position is expressed in the base frame; in $r_{c,EE}$ the index after the comma identifies the coordinate frame the components are expressed in.

2.1 Reference Frames

The theoretical formulation and controller use these right-handed coordinate frames [19, 3].

- **Base frame $\{0\}$:** The base frame is attached to the robot mounting surface. Its origin is the robot base reference point used by the kinematic model. For the upright Panda configuration used in this thesis, the z_0 -axis is normal to the mounting

surface, while the x_0 - and y_0 -axes lie in the mounting plane.

- **Joint frames $\{1\}, \{2\}, \dots, \{7\}$:** The joint frames follow the robot kinematic convention and are required for forward kinematics and Jacobian construction; the real-time controller obtains the corresponding quantities from the libfranka model.
- **End-effector frame $\{EE\}$:** The end-effector frame is attached to the controlled TCP and moves with the tool. In the implemented robot model, the end-effector origin and TCP are identified, so $p_{TCP} = p_{EE}$. The physical tool face is represented separately by its calibrated geometric offset from this frame. The position p_{EE} and orientation R_{EE} , both expressed relative to the base frame, define the current Cartesian pose of the robot. The axes x_{EE} , y_{EE} , and z_{EE} describe the current end-effector orientation. The rotation R_{EE} maps components from end-effector coordinates into base coordinates, so a tool direction $n_{Tool,EE}$ calibrated in this frame has the base-frame representation $n_{Tool} = R_{EE}n_{Tool,EE}$.
- **Desired end-effector frame $\{d\}$:** The desired frame represents the commanded TCP pose. Its origin is p_d , and its orientation is R_d , both expressed relative to the base frame. The difference between the desired and current end-effector frames provides the translational and rotational errors used by the impedance controller.
- **Surface frame $\{S\}$:** The surface frame is attached to the configured surface and remains stationary as the tool moves. Its origin is placed at the configured surface point p_s . Its orientation $R_{surface}$ is determined by the two surface tangents t_1 and t_2 and the surface normal n_s . Its construction is given in Section 2.5.

The directions t_1 and t_2 lie in the surface plane and are mutually orthogonal. The unit normal n_s is defined to point outward from the surface towards the free-space region above the plate. Motion towards the surface follows $-n_s$ throughout the contact sequence. The surface frame is used to describe the configured orientation offset and to assign stiffness and damping values according to the two tangential directions and the normal direction.

Figure 2.1 shows these frames together with the position error between the current and the desired end-effector frame.

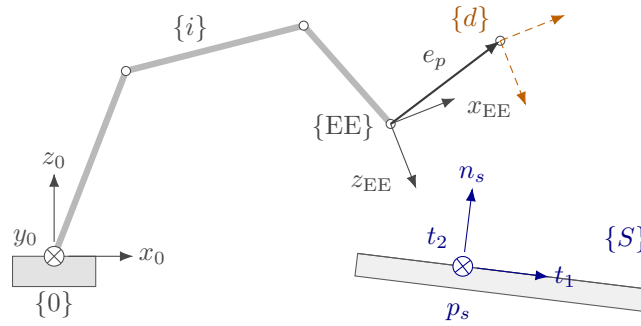


Figure 2.1: Frames used in the controller and the end-effector position error.

2.2 Rigid-Body Pose and Pose Error

Unless stated otherwise, Cartesian quantities are expressed in the robot base frame. A Cartesian pose consists of a position $p \in \mathbb{R}^3$ and an orientation $R \in SO(3)$, collected in the homogeneous transformation T :

$$T = \begin{bmatrix} R & p \\ 0^\top & 1 \end{bmatrix}. \quad (2.1)$$

The measured end-effector pose is

$$T_{\text{EE}} = \begin{bmatrix} R_{\text{EE}} & p_{\text{EE}} \\ 0^\top & 1 \end{bmatrix}. \quad (2.2)$$

The controller obtains T_{EE} directly from the libfranka robot state and uses R_{EE} for orientation calculations.

2.2.1 Cartesian Pose Error

The desired end-effector transformation T_d is

$$T_d = \begin{bmatrix} R_d & p_d \\ 0^\top & 1 \end{bmatrix}, \quad (2.3)$$

where p_d and R_d are the desired TCP position and orientation, respectively. The current end-effector pose is denoted by T_{EE} .

The relative pose-error transformation T_{err} maps the current end-effector frame to the desired frame:

$$T_{\text{err}} = T_{\text{EE}}^{-1} T_d = \begin{bmatrix} R_{\text{EE}}^\top R_d & R_{\text{EE}}^\top (p_d - p_{\text{EE}}) \\ 0^\top & 1 \end{bmatrix}. \quad (2.4)$$

The translational block of Equation (2.4) gives the position difference in the current end-effector frame. For the Cartesian impedance law, the position error e_p is expressed in the base frame:

$$e_p = p_d - p_{\text{EE}} \in \mathbb{R}^3. \quad (2.5)$$

The position error e_p points from the current TCP position towards the desired TCP position.

The rotational block of T_{err} gives the orientation difference in the current end-effector frame. Expressing this rotation in the base frame gives ΔR :

$$\Delta R = R_d R_{\text{EE}}^\top. \quad (2.6)$$

This expression follows by conjugating the current-frame block:

$$R_{\text{EE}} (R_{\text{EE}}^\top R_d) R_{\text{EE}}^\top = R_d R_{\text{EE}}^\top.$$

It satisfies $\Delta R R_{\text{EE}} = R_d$ and therefore represents the rotation that maps the measured orientation to the desired orientation, directly in base coordinates.

The axis-angle form of ΔR consists of a rotation angle ϕ and a unit rotation axis $u \in \mathbb{R}^3$ [19, 16]. The rotation-matrix and axis-angle representations can be converted in both directions. Given a unit axis u and an angle ϕ , the corresponding rotation matrix is obtained from Rodrigues' formula,

$$R(u, \phi) = I + \sin \phi [u]_\times + (1 - \cos \phi) [u]_\times^2. \quad (2.7)$$

For the relative rotation considered here, $R(u, \phi) = \Delta R$. The skew-symmetric matrix of

the rotation axis is

$$[u]_{\times} = \begin{bmatrix} 0 & -u_z & u_y \\ u_z & 0 & -u_x \\ -u_y & u_x & 0 \end{bmatrix}. \quad (2.8)$$

Conversely, when the rotation matrix ΔR is known, the angle follows from

$$\text{tr}(\Delta R) = 1 + 2 \cos \phi, \quad (2.9)$$

This gives

$$\phi = \arccos\left(\frac{\text{tr}(\Delta R) - 1}{2}\right), \quad \phi \in [0, \pi]. \quad (2.10)$$

Subtracting the transpose of ΔR isolates its skew-symmetric part:

$$\frac{1}{2}(\Delta R - \Delta R^{\top}) = \sin \phi [u]_{\times}. \quad (2.11)$$

For $\sin \phi \neq 0$, the rotation axis is obtained from

$$u = \frac{1}{2 \sin \phi} \begin{bmatrix} \Delta R_{32} - \Delta R_{23} \\ \Delta R_{13} - \Delta R_{31} \\ \Delta R_{21} - \Delta R_{12} \end{bmatrix}. \quad (2.12)$$

Multiplying the base-frame unit axis by the angle gives the controller orientation error e_R directly:

$$e_R = \phi u \in \mathbb{R}^3. \quad (2.13)$$

The direction of e_R defines the axis of the required correction, while its magnitude is the angular error in radians. Both e_p and e_R are expressed in the base frame and point in the translational and rotational directions required to reduce the pose error.

The complete Cartesian error e stacks the position error e_p and orientation error e_R used by the impedance controller:

$$e = \begin{bmatrix} e_p \\ e_R \end{bmatrix} \in \mathbb{R}^6. \quad (2.14)$$

2.3 Kinematics of a 7-DOF Manipulator

2.3.1 Forward Kinematics

For $q \in \mathbb{R}^7$, forward kinematics maps the seven joint coordinates to the end-effector pose. Using the joint frames introduced in Section 2.1, the transformation from the base frame $\{0\}$ to link 7 is

$${}^0T_7(q) = \prod_{i=1}^7 {}^{i-1}T_i(q_i). \quad (2.15)$$

In the standard Denavit–Hartenberg convention [4, 3, 19], joint i rotates by q_i about the axis z_{i-1} . The transformation between two consecutive joint frames is

$${}^{i-1}T_i(q_i) = R_z(q_i) T_z(d_i) T_x(a_i) R_x(\alpha_i), \quad (2.16)$$

where a_i , d_i , and α_i describe the fixed geometry between consecutive joint frames. The transformation ${}^7T_{EE}$ locates the configured end-effector frame relative to link 7. The complete base-to-end-effector transformation is

$$T_{EE}(q) = {}^0T_7(q) {}^7T_{EE} = \begin{bmatrix} R_{EE}(q) & p_{EE}(q) \\ 0^\top & 1 \end{bmatrix}. \quad (2.17)$$

The controller receives T_{EE} and the geometric Jacobian from the libfranka model interface. The geometric Jacobian $J(q) \in \mathbb{R}^{6 \times 7}$ maps joint velocity to the base-frame end-effector twist,

$$\begin{bmatrix} \dot{p}_{EE} \\ \omega_{EE} \end{bmatrix} = J(q) \dot{q}. \quad (2.18)$$

Its transpose maps the commanded Cartesian wrench to joint torque. These two mappings are the kinematic relations required by the impedance and null-space controllers.

2.4 Cartesian Impedance Control

The controller evaluates a base-frame Cartesian spring–damper wrench F from the pose error e , stiffness K , damping D , and the measured linear and angular velocities [11, 18, 14]. The geometric Jacobian transpose maps this wrench to joint torque $\tau_{\text{cart}} = J^\top F$ [12].

The quantities F , f , and m denote commanded controller quantities throughout this chapter. They are separate from the model-estimated external wrench that libfranka supplies and the experiments record.

2.4.1 Impedance Wrench

The impedance wrench is evaluated in the robot base frame. The corresponding translational stiffness and damping matrices are $K_p, D_p \in \mathbb{R}^{3 \times 3}$, while the rotational matrices are $K_R, D_R \in \mathbb{R}^{3 \times 3}$. Rotating the diagonal surface-frame stiffness and damping matrices into the base frame generally produces non-zero off-diagonal entries. Their base-frame representation changes with the surface-frame orientation, even when the local matrix entries remain unchanged.

The commanded Cartesian force f is given by the translational spring–damper law [11, 14]

$$f = K_p e_p + D_p (\dot{p}_d - \dot{p}_{\text{EE}}) \in \mathbb{R}^3, \quad (2.19)$$

where $\dot{p}_d$ and $\dot{p}_{\text{EE}}$ are the desired and measured linear velocities expressed in the base frame. For a fixed position setpoint, $\dot{p}_d = \mathbf{0}$, and the damping term becomes $-D_p \dot{p}_{\text{EE}}$.

The commanded Cartesian moment m is

$$m = K_R e_R + D_R (\omega_d - \omega_{\text{EE}}) \in \mathbb{R}^3, \quad (2.20)$$

where ω_d and ω_{EE} are the desired and measured end-effector angular velocities expressed in the base frame.

The Cartesian wrench F stacks the force and moment:

$$F = \begin{bmatrix} f \\ m \end{bmatrix} = \begin{bmatrix} K_p e_p + D_p (\dot{p}_d - \dot{p}_{EE}) \\ K_R e_R + D_R (\omega_d - \omega_{EE}) \end{bmatrix} \in \mathbb{R}^6. \quad (2.21)$$

The wrench is ordered as force followed by moment, and all its components are expressed in the base frame. The Jacobian transpose maps it to the joint torques τ_{cart} .

Equation (2.21) defines the impedance about the TCP, with translation and rotation decoupled. Section 2.7 extends the same law to an impedance defined about a displaced centre of compliance.

2.4.2 Joint-Torque Control Law

By the principle of virtual work, the commanded Cartesian wrench F is mapped to joint torques τ_{cart} through the transpose of the geometric Jacobian $J(q)$ [12]:

$$\tau_{\text{cart}} = J^\top(q)F, \quad (2.22)$$

where $\tau_{\text{cart}} \in \mathbb{R}^7$ is the Cartesian impedance torque contribution and $J^\top(q) \in \mathbb{R}^{7 \times 6}$. Substituting Equation (2.21) gives

$$\tau_{\text{cart}} = J^\top(q) \begin{bmatrix} K_p e_p + D_p (\dot{p}_d - \dot{p}_{EE}) \\ K_R e_R + D_R (\omega_d - \omega_{EE}) \end{bmatrix}. \quad (2.23)$$

The stiffness and damping matrices in this expression are obtained from their surface-coordinate representations. A change in the surface-frame orientation changes the Cartesian wrench F and the resulting joint-torque contribution τ_{cart} , even when the local stiffness and damping values remain unchanged.

2.5 Surface Frame

The surface frame provides the tangential and normal directions used for the Cartesian controller quantities. Its orientation is constructed from the unit surface normal n_s , expressed in the robot base frame. The sign of n_s is chosen so that it points outward from the surface towards the free-space region above the plate, which makes $-n_s$ the approach

and contact-establishment direction.

The first surface tangent is chosen to follow the base-frame x_0 -direction as closely as possible while remaining in the surface plane. The configured surface is inclined relative to the base x_0y_0 -plane, so e_{x_0} is projected onto the surface plane and normalised:

$$t_1 = \frac{e_{x_0} - n_s (n_s^\top e_{x_0})}{\|e_{x_0} - n_s (n_s^\top e_{x_0})\|}. \quad (2.24)$$

The second tangent completes the right-handed frame,

$$t_2 = n_s \times t_1. \quad (2.25)$$

The construction applies for $|e_{x_0}^\top n_s| < 1$, and the resulting t_1 and t_2 are orthogonal unit vectors in the surface plane. A different in-plane reference rotates both tangents while preserving the surface geometry. Axis-specific results refer to the configured direction, whose configured in-plane reference is recorded in Appendix C. The tangents obtained for the contact experiments are given in Section 4.2.

The orientation of the surface frame relative to the robot base frame is

$$R_{\text{surface}} = \begin{bmatrix} t_1 & t_2 & n_s \end{bmatrix}. \quad (2.26)$$

Surface-frame quantities are listed in the column order of Equation (2.26), that is t_1 , then t_2 , then n_s .

The quantities K_{p,t_1} , K_{p,t_2} , and $K_{p,n}$ denote the translational stiffness entries along the two surface tangents and the surface normal, respectively. These are scalar directional coefficients, and their direction is already specified by t_1 , t_2 , and n_s , so they carry no frame suffix. The same ordering is used for the rotational stiffness and damping and for the surface-frame components of the compliance-centre displacement.

2.6 Stiffness and Damping Representation

The Cartesian impedance law of Section 2.4 is evaluated in the robot base frame. The directional stiffness and damping values are specified relative to the configured surface

reference, because the relevant directions for the contact problem are the two surface tangents t_1 and t_2 and the surface normal n_s .

The translational and rotational stiffness and damping matrices are first defined in surface coordinates as

$$K_{p,S} = \begin{bmatrix} K_{p,t_1} & 0 & 0 \\ 0 & K_{p,t_2} & 0 \\ 0 & 0 & K_{p,n} \end{bmatrix}, \quad D_{p,S} = \begin{bmatrix} D_{p,t_1} & 0 & 0 \\ 0 & D_{p,t_2} & 0 \\ 0 & 0 & D_{p,n} \end{bmatrix}, \quad (2.27)$$

$$K_{R,S} = \begin{bmatrix} K_{R,t_1} & 0 & 0 \\ 0 & K_{R,t_2} & 0 \\ 0 & 0 & K_{R,n} \end{bmatrix}, \quad D_{R,S} = \begin{bmatrix} D_{R,t_1} & 0 & 0 \\ 0 & D_{R,t_2} & 0 \\ 0 & 0 & D_{R,n} \end{bmatrix}. \quad (2.28)$$

The common axis order is $[t_1, t_2, n_s]$.

The matrices $K_{p,S}$, $D_{p,S}$, $K_{R,S}$, and $D_{R,S}$ hold the directional gains in surface coordinates. Rotating them with R_{surface} gives the base-frame matrices K_p , D_p , K_R , and D_R used by the impedance law:

$$K_p = R_{\text{surface}} K_{p,S} R_{\text{surface}}^\top, \quad D_p = R_{\text{surface}} D_{p,S} R_{\text{surface}}^\top, \quad (2.29)$$

$$K_R = R_{\text{surface}} K_{R,S} R_{\text{surface}}^\top, \quad D_R = R_{\text{surface}} D_{R,S} R_{\text{surface}}^\top. \quad (2.30)$$

The complete base-frame stiffness and damping matrices are then

$$K = \begin{bmatrix} K_p & 0 \\ 0 & K_R \end{bmatrix}, \quad D = \begin{bmatrix} D_p & 0 \\ 0 & D_R \end{bmatrix}. \quad (2.31)$$

The transformation rotates the matrix representation into base coordinates while preserving the physical directions defined by the configured surface reference. The normal entry $K_{p,n}$ therefore follows an inclined n_s relative to the robot-base axes.

At this stage translation and rotation remain decoupled. The centre-of-compliance transformation introduced in Section 2.7 can subsequently introduce coupling between the two blocks.

2.6.1 Directional Cartesian Inertia and Inertia-Scaled Damping

The damping coefficients are calculated from the stiffness and the Cartesian inertia along the same direction. The regularised operational-space inertia $\Lambda(q)$ is

$$\Lambda(q) = \left(J(q)M^{-1}(q)J^\top(q) + \varepsilon I_6 \right)^{-1}, \quad (2.32)$$

where $M(q)$ is the joint-space inertia and $\varepsilon > 0$ is a regularisation constant. The block rotation $T_R = \text{diag}(R_{\text{surface}}, R_{\text{surface}})$ resolves this matrix along the surface-frame directions. The resulting surface-resolved operational-space inertia $\Lambda_S(q)$ and its directional entries $\lambda_i(q)$ are

$$\Lambda_S(q) = T_R^\top \Lambda(q) T_R, \quad \lambda_i(q) = [\Lambda_S(q)]_{ii}. \quad (2.33)$$

The index i denotes one translational or rotational surface-frame direction. Each controller state or configuration supplies the active stiffness entry k_i for that direction. The corresponding inertia-scaled damping coefficient $d_i(q)$ is

$$d_i(q) = 2\zeta \sqrt{\lambda_i(q)k_i}, \quad (2.34)$$

where ζ is the selected damping factor. For one decoupled direction, $\zeta = 1$ gives the critical-damping coefficient of the corresponding local second-order model.

2.7 Centre of Compliance

The surface-frame transformations of Equations (2.29) and (2.30) change the directions along which the Cartesian stiffness and damping act, while leaving their reference point at the tool centre point (TCP). Independently of that orientation choice, the Cartesian impedance can be defined about a virtual CoC p_c . Shifting this point away from the TCP introduces translation–rotation coupling into the Cartesian impedance [16, 5, 20, 13]. The construction is applied during Contact Establishment, where it lets the normal pressing action generate a rotational moment as well as a translational contact load.

The displacement from the TCP to the CoC is

$$r_c = p_c - p_{\text{TCP}} \in \mathbb{R}^3, \quad p_{\text{TCP}} = p_{\text{EE}}. \quad (2.35)$$

The vector r_c is the virtual shift of the impedance reference point.

Let $[r_c]_\times$ denote the skew-symmetric matrix satisfying $[r_c]_\times a = r_c \times a$. Under the local small-angle formulation, the translational error referred to the CoC is

$$e_{p,c} = e_p - [r_c]_\times e_R. \quad (2.36)$$

The negative sign follows from the selected displacement convention. The vector from the TCP to p_c is r_c itself, giving

$$e_R \times r_c = -[r_c]_\times e_R.$$

The stacked error at the CoC collects $e_{p,c}$ and the rotational error e_R . The point-shift adjoint $\text{Ad}(r_c)$ maps the corresponding TCP error onto it:

$$e_c = \begin{bmatrix} e_{p,c} \\ e_R \end{bmatrix} = \text{Ad}(r_c) \begin{bmatrix} e_p \\ e_R \end{bmatrix}, \quad \text{Ad}(r_c) = \begin{bmatrix} I_3 & -[r_c]_\times \\ 0 & I_3 \end{bmatrix}. \quad (2.37)$$

The surface-frame rotation and the point shift have different functions. The transformations in Equations (2.29) and (2.30) change the principal directions of the impedance. The point-shift adjoint $\text{Ad}(r_c)$ changes the point about which the impedance is defined.

After the stiffness and damping have been expressed in the base frame, the translational and rotational matrices are assembled at the CoC as follows. The index c identifies the reference point of these matrices; their components remain expressed in the base frame.

$$K_c = \begin{bmatrix} K_p & 0 \\ 0 & K_R \end{bmatrix}, \quad D_c = \begin{bmatrix} D_p & 0 \\ 0 & D_R \end{bmatrix}. \quad (2.38)$$

All blocks in Equation (2.38) are expressed in the base-frame orientation. The point-shift transformation is applied to both the stiffness K_c and damping D_c . The corresponding matrices at the TCP follow as

$$K_{\text{TCP}} = \text{Ad}(r_c)^\top K_c \text{Ad}(r_c) = \begin{bmatrix} K_p & -K_p[r_c]_\times \\ -[r_c]_\times^\top K_p & K_R + [r_c]_\times^\top K_p[r_c]_\times \end{bmatrix} \quad (2.39)$$

and

$$D_{\text{TCP}} = \text{Ad}(r_c)^\top D_c \text{Ad}(r_c) = \begin{bmatrix} D_p & -D_p[r_c]_\times \\ -[r_c]_\times^\top D_p & D_R + [r_c]_\times^\top D_p[r_c]_\times \end{bmatrix}. \quad (2.40)$$

The off-diagonal blocks introduce translation–rotation coupling. A translational position or velocity error can contribute to the commanded moment, while a rotational error or velocity can contribute to the commanded force. For $r_c = 0$ the point shift disappears and the block-diagonal impedance at the TCP is recovered.

2.7.1 Commanded Moment

The Cartesian wrench generated by the impedance controller is $F = [f^\top, m^\top]^\top$, where f and m denote the complete commanded force and moment at the TCP.

At $r_c = 0$, the ordinary rotational impedance gives the commanded moment,

$$m_R = K_R e_R - D_R \omega_{EE}. \quad (2.41)$$

A non-zero CoC displacement adds the translation–rotation coupling contribution $r_c \times f$, generated by the complete commanded force. The complete commanded TCP moment is

$$m = m_R + r_c \times f. \quad (2.42)$$

The shifted CoC makes the commanded press act as if it had a virtual lever arm r_c relative to the TCP, which generates the additional moment $r_c \times f$.

Equation (2.42) is the analytical decomposition of the complete shifted spring–damper wrench evaluated by the controller.

For $r_c = 0$ the coupling contribution vanishes and $m = m_R$. The TCP-centred condition removes the coupling contribution while the finite rotational compliance and commanded rotational impedance remain active.

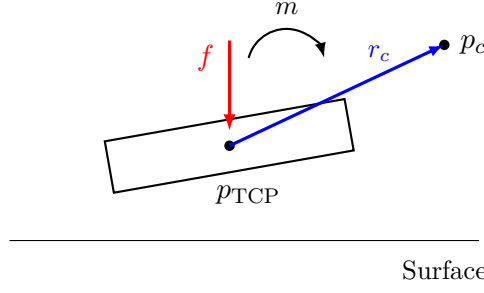


Figure 2.2: Compliance-centre displacement and commanded TCP moment.

2.7.2 Tangential Compliance-Centre Displacement

For surface contact the CoC displacement is resolved along the two surface tangents and the surface normal. Its coordinates in the surface frame are

$$r_{c,S} = \begin{bmatrix} r_{c,t_1} \\ r_{c,t_2} \\ r_{c,n} \end{bmatrix}. \quad (2.43)$$

Using the surface-frame transformation, the same physical displacement in base coordinates is

$$r_c = R_{\text{surface}} r_{c,S} = r_{c,t_1} t_1 + r_{c,t_2} t_2 + r_{c,n} n_s, \quad (2.44)$$

with the inverse relation $r_{c,S} = R_{\text{surface}}^\top r_c$. The tangential part collects the two in-plane components,

$$r_{c,t} = r_{c,t_1} t_1 + r_{c,t_2} t_2, \quad (2.45)$$

while $r_{c,n} n_s$ is the part along the surface normal. The scalar coordinates r_{c,t_1} and r_{c,t_2} may be positive or negative. Their tangential vector is $r_{c,t}$, with magnitude $\|r_{c,t}\|$. For a purely tangential displacement $r_{c,n} = 0$, giving $\|r_c\| = \|r_{c,t}\|$.

A non-zero displacement is specified by the frame in which its components are held constant as well as by its magnitude and direction. The surface-fixed definition is given by Equation (2.44). For a tool-fixed definition, the end-effector orientation carries the configured vector into the base frame:

$$r_c = R_{EE}r_{c,EE}. \quad (2.46)$$

For a surface-fixed definition $r_{c,S}$ remains constant, and because the surface frame does not change during one controller session the base-frame displacement is constant during contact establishment. For a tool-fixed definition $r_{c,EE}$ remains constant, so r_c rotates with the end effector. Section 3.2.5 describes both implementations.

The commanded force enters through $r_c \times f$, and it decomposes exactly into a normal and a tangential component,

$$f = f_n + f_t, \quad f_n = F_n n_s, \quad f_t = f - f_n, \quad (2.47)$$

with $F_n = n_s^\top f$ the commanded normal-force component. The normal component is the projection of f onto n_s , and the tangential component is what remains of f in the surface plane. The complete CoC contribution is

$$r_c \times f = r_c \times f_n + r_c \times f_t. \quad (2.48)$$

Both terms remain part of the complete commanded moment m .

The normal-press term simplifies. Substituting $r_c = r_{c,t} + r_{c,n}n_s$ and using $f_n \parallel n_s$, the normal component of the displacement drops out, $r_{c,n}n_s \times f_n = \mathbf{0}$, which leaves

$$r_c \times f_n = r_{c,t} \times f_n. \quad (2.49)$$

The commanded normal press couples through the tangential projection $r_{c,t}$, independently of the normal component $r_{c,n}$.

The right-handed surface frame satisfies

$$t_1 \times t_2 = n_s, \quad n_s \times t_1 = t_2, \quad n_s \times t_2 = -t_1. \quad (2.50)$$

Figure 2.3 shows the construction in three views. Panel (a) is an oblique view carrying the two rotations about the surface tangents. Panel (b) is a view along n_s , in which their first-order contributions combine into one direction in the surface tangent plane. Panel (c)

carries the displacement selected perpendicular to that direction.

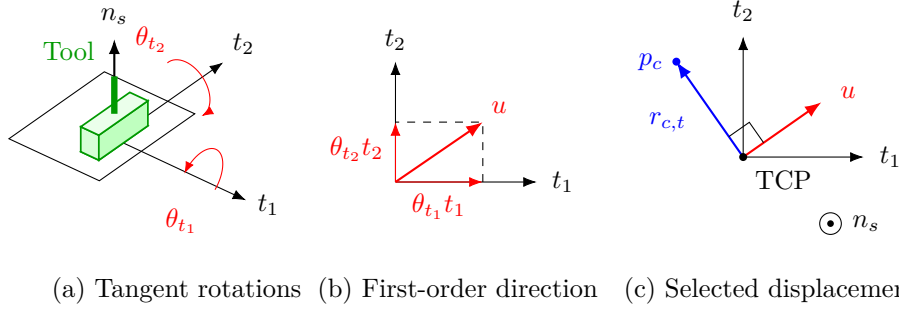


Figure 2.3: Tangent rotations, their combined direction, and the selected displacement.

An angular offset of the tool can contain rotations θ_{t_1} and θ_{t_2} about the two surface tangents. For a unit rotation axis t_i , rotating any vector v by θ_{t_i} gives exactly

$$R(t_i, \theta_{t_i})v = v \cos \theta_{t_i} + (t_i \times v) \sin \theta_{t_i} + t_i(t_i^\top v)(1 - \cos \theta_{t_i}), \quad (2.51)$$

which is Rodrigues' rotation formula of Equation (2.7) written directly with vectors rather than with $[t_i]_\times$.

If the rotation about t_1 is applied first and the rotation about t_2 second, the exact composition is

$$v' = R(t_2, \theta_{t_2})R(t_1, \theta_{t_1})v. \quad (2.52)$$

For small angular offsets,

$$\sin \theta_{t_i} \approx \theta_{t_i}, \quad \cos \theta_{t_i} \approx 1, \quad 1 - \cos \theta_{t_i} \approx 0$$

to first order. Therefore Rodrigues' formula reduces to

$$R(t_i, \theta_{t_i})v \approx v + \theta_{t_i}(t_i \times v).$$

After the first rotation,

$$v_1 \approx v + \theta_{t_1}(t_1 \times v),$$

and after the second,

$$v' \approx v_1 + \theta_{t_2}(t_2 \times v_1).$$

Substituting v_1 ,

$$v' \approx v + \theta_{t_1}(t_1 \times v) + \theta_{t_2}(t_2 \times v) + \theta_{t_1}\theta_{t_2}[t_2 \times (t_1 \times v)].$$

The final term contains the product $\theta_{t_1}\theta_{t_2}$, so it is second order and is neglected in the small-angle approximation. This leaves

$$v' \approx v + (\theta_{t_1}t_1 + \theta_{t_2}t_2) \times v. \quad (2.53)$$

The exact composition depends on the order of the two finite rotations, whereas this difference enters only through second- and higher-order terms and therefore disappears in the first-order small-angle representation.

Thus, to first order, the two small angular contributions define one direction in the surface tangent plane. For a non-zero angular offset, the corresponding unit direction is

$$u = \frac{\theta_{t_1}t_1 + \theta_{t_2}t_2}{\sqrt{\theta_{t_1}^2 + \theta_{t_2}^2}}. \quad (2.54)$$

For a pure rotation about t_1 , $u = t_1$, and for a pure rotation about t_2 , $u = t_2$. When both angular offsets are present, u gives their combined first-order direction in the t_1 - t_2 plane. Reversing the signs of both angular offsets reverses u .

The tangential CoC displacement is selected so that the moment generated by the commanded normal force supports reduction of this angular offset. The normal commanded force is

$$f_n = F_n n_s,$$

with $F_n < 0$ during Contact Establishment. The desired CoC-induced moment must therefore point opposite to u . Since

$$r_{c,t} \times f_n = F_n(r_{c,t} \times n_s),$$

and $F_n < 0$, the required condition is that

$$r_{c,t} \times n_s$$

points along u .

Because $u \perp n_s$ and both vectors have unit magnitude,

$$(n_s \times u) \times n_s = u.$$

The tangential CoC displacement is therefore selected as

$$r_{c,t} = \|r_{c,t}\|(n_s \times u). \quad (2.55)$$

Substituting Equation (2.54) and using Equation (2.50) gives the explicit relation

$$r_{c,t} = \|r_{c,t}\| \frac{\theta_{t_1} t_2 - \theta_{t_2} t_1}{\sqrt{\theta_{t_1}^2 + \theta_{t_2}^2}}. \quad (2.56)$$

The magnitude $\|r_{c,t}\|$ remains a selectable controller parameter, while Equation (2.56) determines its tangential direction from the two small angular offsets.

The corresponding normal-force contribution to the commanded moment is

$$\begin{aligned} r_{c,t} \times f_n &= \|r_{c,t}\|(n_s \times u) \times (F_n n_s) \\ &= \|r_{c,t}\| F_n u. \end{aligned} \quad (2.57)$$

Since $F_n < 0$, this moment points opposite to u and therefore supports reduction of the angular offset.

For the two principal directions, Equation (2.55) reduces to

$$r_{c,t} = +\|r_{c,t}\|t_2 \quad \text{for } u = t_1, \quad r_{c,t} = -\|r_{c,t}\|t_1 \quad \text{for } u = t_2. \quad (2.58)$$

Reversing the angular-offset direction reverses u , $r_{c,t}$, and the resulting CoC-induced moment. Figure 2.4 depicts the selected displacement in panel (a) and its reversal in panel (b). For $\theta_{t_1} = \theta_{t_2} = 0$, no preferred direction u exists and the directional CoC-selection rule is not applied.

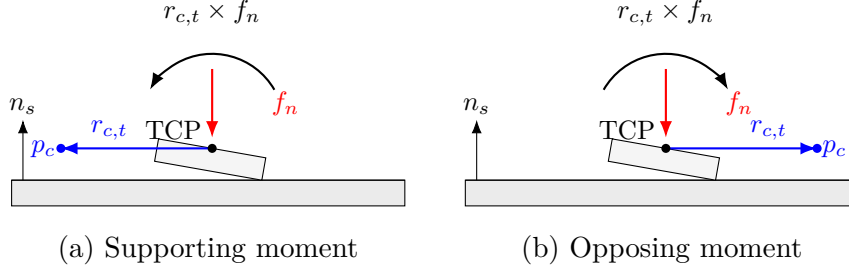


Figure 2.4: Normal-force contribution for (a) the selected tangential displacement and (b) its reversal.

The direction rule isolates $r_c \times f_n$; the tangential-force contribution remains in the complete commanded moment of Equation (2.42).

The coupling vanishes for $r_c = 0$. Conversely, if K_p is nonsingular, zero coupling implies $r_c = 0$. Since $\text{Ad}(r_c)$ is invertible, the point-shift transformation also preserves definiteness:

$$K_c \succeq 0 \implies K_{\text{TCP}} \succeq 0, \quad K_c \succ 0 \implies K_{\text{TCP}} \succ 0. \quad (2.59)$$

The same statements apply to D_c and D_{TCP} . The negative off-diagonal entries are coupling terms within the positive semidefinite complete matrices.

This formulation is a local small-motion model in which the displacement is treated as constant during one controller evaluation. A surface-fixed $r_{c,S}$ keeps r_c constant during contact establishment. A tool-fixed displacement rotates with the end effector, so the controller updates the base-frame vector and the shifted stiffness and damping matrices during each control cycle.

The virtual CoC p_c is the selected impedance reference. The physical contact location can move across the tool face.

2.8 Null-Space Torque

The Panda has one redundant joint direction when its 6×7 geometric Jacobian $J(q)$ has full rank. The singular value decomposition (SVD) supplies two different quantities: v_7 spans the redundant direction, while $\sigma_{\min} = \sigma_6$ indicates the weakest task-producing mapping. A thresholded pseudoinverse gives the velocity and torque projectors. The controller then combines projected joint damping with a two-probe conditioning torque.

2.8.1 SVD and Null-Space Direction

The decomposition of the current Jacobian is

$$J(q) = U_J \Sigma_J V_J^\top, \quad (2.60)$$

where the six task-producing singular values satisfy $\sigma_1 \geq \dots \geq \sigma_6 \geq 0$. At full rank, the seventh right-singular vector satisfies

$$Jv_7 = 0, \quad (2.61)$$

and spans the one-dimensional kinematic null space. The conditioning indicator is

$$\sigma_{\min} = \sigma_6. \quad (2.62)$$

The vector v_7 and scalar σ_6 have different roles. The first defines redundant motion; the second describes the weakest of the six task mappings. Because the geometric Jacobian mixes linear and angular rows, $\sigma_{\min}$ serves as a relative conditioning indicator under the fixed Jacobian convention.

2.8.2 Pseudoinverse and Projectors

A relative tolerance ρ defines the retained singular values by $\sigma_i > \rho\sigma_1$. The corresponding Moore–Penrose pseudoinverse is

$$J^+ = \sum_{\sigma_i > \rho\sigma_1}^6 \frac{1}{\sigma_i} v_i u_i^\top. \quad (2.63)$$

The joint-velocity projector extracts redundant motion,

$$N_q = I_7 - J^+ J, \quad \dot{q}_{\text{null}} = N_q \dot{q}, \quad (2.64)$$

while the transpose projector confines a secondary joint torque to the kinematic null space,

$$N_\tau = N_q^\top = I_7 - J^\top J^{+\top}. \quad (2.65)$$

With the thresholded SVD construction, the projector is orthogonal and symmetric. At full rank it reduces to $N_q = N_\tau = v_7 v_7^\top$. The implementation uses this kinematic projector. Accordingly, the pose-hold experiment evaluates Cartesian position retention.

2.8.3 Projected Damping and Conditioning

Projected null-space damping opposes the redundant joint velocity,

$$\tau_d = -d_{\text{null}} N_\tau \dot{q}. \quad (2.66)$$

The conditioning term samples two configurations along the current redundant direction. For the probe angle α_{probe} , these are

$$q_+ = q + \alpha_{\text{probe}} v_7, \quad q_- = q - \alpha_{\text{probe}} v_7. \quad (2.67)$$

Their minimum singular values are

$$\sigma_+ = \sigma_{\min}(J(q_+)), \quad \sigma_- = \sigma_{\min}(J(q_-)), \quad (2.68)$$

with the sampled difference

$$\Delta_\sigma = \sigma_+ - \sigma_-. \quad (2.69)$$

The direction selector applies a deadband δ_σ ,

$$s_\sigma = \begin{cases} +1, & \Delta_\sigma > \delta_\sigma, \\ -1, & \Delta_\sigma < -\delta_\sigma, \\ 0, & |\Delta_\sigma| \leq \delta_\sigma, \end{cases} \quad (2.70)$$

and the conditioning torque is

$$\tau_\sigma = k_\sigma N_\tau(s_\sigma v_7). \quad (2.71)$$

Sampling q_+ and q_- selects the neighbouring configuration with the larger $\sigma_{\min}$. The finite probes are local tangent approximations to the nonlinear self-motion. The commanded magnitude k_σ is independent of the probe angle.

2.8.4 Complete Null-Space Torque

The complete secondary torque is

$$\tau_{\text{null}} = \tau_d + \tau_\sigma = -d_{\text{null}} N_\tau \dot{q} + k_\sigma N_\tau(s_\sigma v_7). \quad (2.72)$$

The damping term is proportional to redundant joint velocity and vanishes at zero velocity. The conditioning term is an active configuration objective and can act at zero joint velocity. Its direction is invariant to the arbitrary sign returned for v_7 , because reversing v_7 also exchanges the two probes and reverses s_σ .

2.9 Coriolis and Gravity Compensation in libfranka

In external joint-torque mode, the Franka Control Interface compensates gravity and motor friction internally [7]. The model-based term the controller supplies is therefore the Coriolis and centrifugal vector $\tau_c(q, \dot{q})$, which libfranka evaluates from the robot model at the measured joint position and velocity.

The vector τ_c depends on the joint state alone, so it is independent of the Cartesian reference and of the secondary null-space torque. It enters the command as a separate

additive contribution.

2.10 Complete Joint-Torque Command

The complete joint-torque command τ_{cmd} combines the Cartesian impedance torque τ_{cart} , the null-space torque τ_{null} , and the Coriolis compensation τ_c :

$$\tau_{\text{cmd}} = \tau_{\text{cart}} + \tau_{\text{null}} + \tau_c(q, \dot{q}) = J^\top(q)F + \tau_d + \tau_\sigma + \tau_c(q, \dot{q}). \quad (2.73)$$

Here $\tau_{\text{cart}} = J^\top(q)F$ is the Cartesian impedance torque. The secondary null-space torque enters as its two contributions, the projected damping τ_d and the singular-value conditioning torque τ_σ of Equation (2.72). The Cartesian and Coriolis terms of this command follow the official libfranka Cartesian impedance example [8], and the null-space torque is added to them.

Chapter 3

Controller Design and Real-Time Implementation

The controller was implemented in C++, using libfranka for communication with the Franka Control Interface and Eigen for matrix operations. At the nominal control period of 1 ms, each callback evaluates the Cartesian impedance wrench, the optional virtual CoC transformation, and the null-space torque.

3.1 Controller Architecture and Real-Time Control Cycle

The software separates session preparation, real-time control, and post-processing. Before a controller session, the program reads and validates the complete parameter set, establishes the FCI connection, requests robot error recovery, applies the selected collision behaviour, loads the robot model, and allocates the logging buffer. During torque control, the real-time callback operates only on the prepared configuration, preallocated state, robot-state data, model quantities, and non-blocking operator requests. File output and experiment evaluation follow after torque control has stopped.

A separate keyboard thread receives operator requests and transfers them through atomic variables, so terminal input stays outside the real-time callback. The parameter file is read again before each session, which permits controller settings to be changed between

experiments.

The operating-state flow begins with robot recovery and controller selection, as shown in Figure 3.1. A successful recovery opens the selection state. Inputs $\mathbf{s}$, $\mathbf{h}$, $\mathbf{t}$, and $\mathbf{g}$ select the Surface-Contact Sequence, Cartesian Pose Hold, Contact-Impedance Hold, and Manual Guidance, respectively. An unsuccessful recovery ends the program. Operator stop, a configured experiment-duration limit, a robot-side error or reflex, and a communication or control exception provide the common transition from any active controller state to Stop.

Libfranka reports robot-side errors and reflexes across its monitored conditions. By contrast, communication and control exceptions form the application-side category.

Contact-Impedance Hold retains the captured Cartesian pose with the impedance used for Contact Establishment. Its connection to the surface-contact sequence is shown in Figure 3.3.

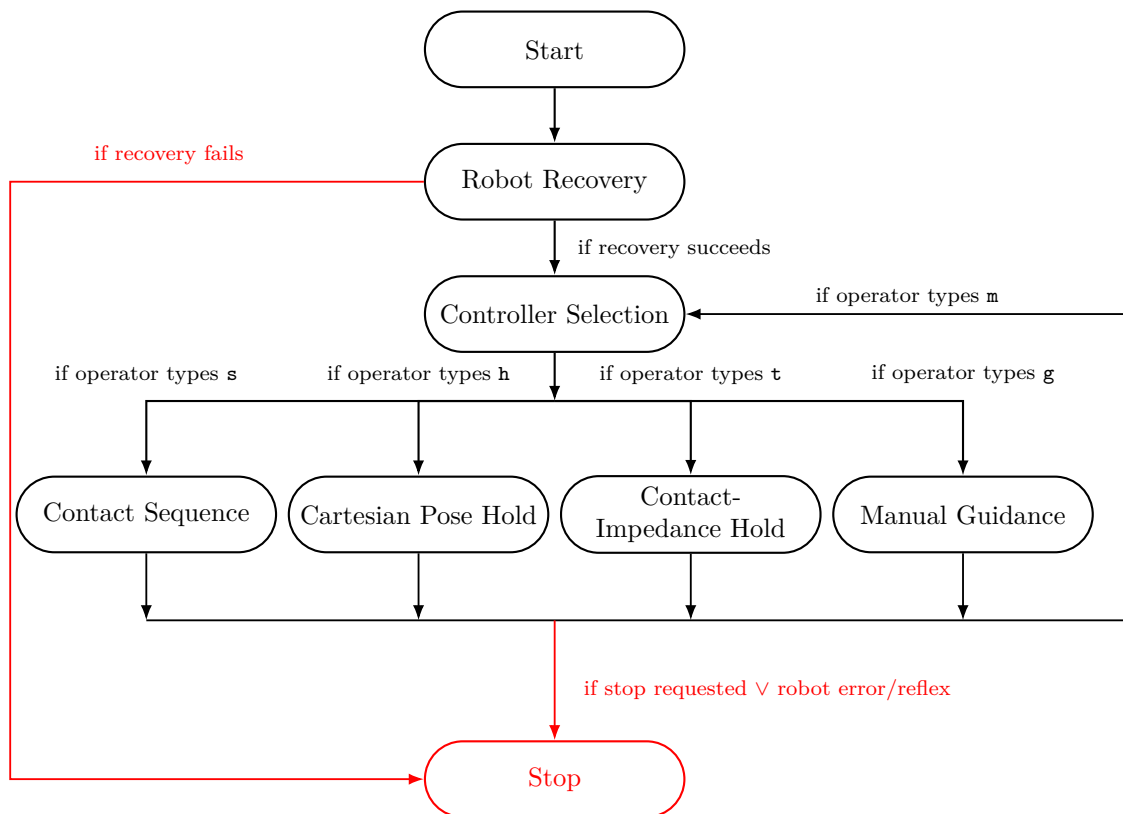


Figure 3.1: Principal controller operating-state flow.

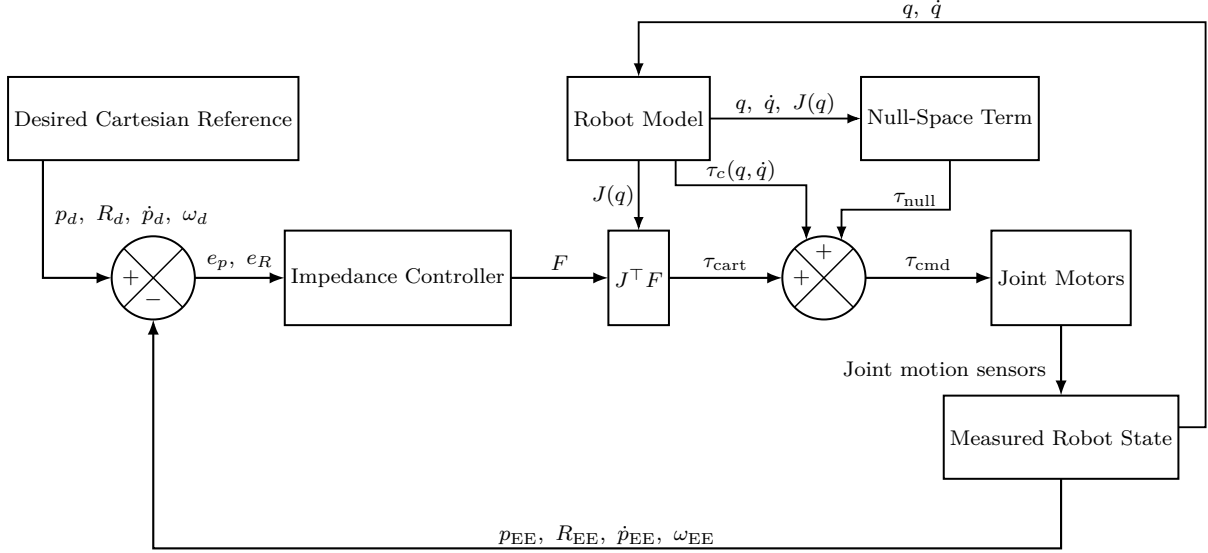


Figure 3.2: Cartesian impedance feedback loop and additive torque terms.

Figure 3.2 shows the computations performed during one real-time control cycle. The active controller state supplies the desired position p_d , orientation R_d , linear velocity $\dot{p}_d$, and angular velocity ω_d . In the implemented controller, $\omega_d = 0$, so the rotational damping term becomes $-D_R \omega_{EE}$. The robot state supplies the measured p_{EE} , R_{EE} , $\dot{p}_{EE}$, and ω_{EE} for feedback.

The Cartesian pose error and the measured velocities determine the impedance wrench $F = [f^\top, m^\top]^\top$ according to Sections 2.2.1 and 2.4. The Jacobian transpose maps this wrench to τ_{cart} . The secondary null-space torque τ_{null} and model Coriolis contribution $\tau_c(q, \dot{q})$ are then added before the commanded joint torque τ_{cmd} is returned to libfranka. During Contact Establishment, the controller keeps R_d fixed at the orientation captured at the clearance transition. Contact-induced end-effector rotation acts against the rotational spring and damper associated with this held reference.

The end-effector pose supplied by the robot state provides p_{EE} and R_{EE} directly.

The rotational error e_R can be resolved in the surface frame so that only the selected rotational components contribute to the rotational spring term. The resulting error is then transformed back to the base frame, while the rotational damping continues to use the measured angular velocity. Depending on the active controller state, the impedance is defined either directly at the TCP or about a displaced virtual CoC. In both cases, the resulting Cartesian wrench is mapped to the joint-torque command through the same torque formulation.

The complete impedance-controlled command follows Equation (2.73).

Here τ_{null} is the secondary null-space torque of Section 2.8.4 and $\tau_c(q, \dot{q})$ is the Coriolis and centrifugal contribution obtained from the libfranka model. The FCI supplies gravity compensation for the external joint-torque command, as described in Section 2.9. The null-space pose-hold experiment of Section 4.7 adds an experiment-specific disturbance torque.

The implemented controller is used in two configurations in this thesis. The surface-contact configuration generates a state-dependent Cartesian reference for Tool Orientation, Surface Approach, Contact Establishment, and the subsequent Grinding state. The Cartesian pose-hold configuration is used for the separate null-space investigation of Section 4.7. Both configurations execute the same real-time control cycle and differ in the generated reference and the active impedance.

3.2 Surface-Contact Sequence

A surface-contact experiment follows the state path drawn in Figure 3.3. It begins either after the robot reaches the stored initial joint configuration q_{init} , or from a configuration reached in Manual Guidance, where input **s** starts the sequence from the current q and input **q** stores that configuration as q_{init} for a later session. The sequence then progresses through Tool Orientation, Surface Approach, the optional Pre-Contact Hold, Contact Establishment, the optional Pre-Grinding Hold, and Grinding.

Tool Orientation ends after its minimum duration, when the configured orientation criteria are satisfied or the orientation timeout is reached. Surface Approach ends when the minimum tool clearance reaches $h_{\text{clearance}}$; reaching the maximum approach distance first terminates the experiment. Each optional hold is skipped when disabled, and otherwise waits for operator confirmation. Contact Establishment ends when its independent timeout is reached.

During active control, operator input **g** transfers the controller to Manual Guidance, where the robot can be repositioned. Input **p** captures the reached pose and returns control according to the previously active mode. For Cartesian Pose Hold and Contact-Impedance Hold, the captured pose becomes the new Cartesian reference, with $p_d = p_{\text{EE}}$

and $R_d = R_{EE}$ at the capture instant. For the surface-contact sequence, the reached configuration provides the starting configuration for restarting the sequence at Tool Orientation.

Operator input $\mathbf{t}$ enters Contact-Impedance Hold from the sequence. That state retains the current Cartesian pose while K_p , K_R , or r_c is set for the next sequence. Input $\mathbf{s}$ then restarts the sequence at Tool Orientation.

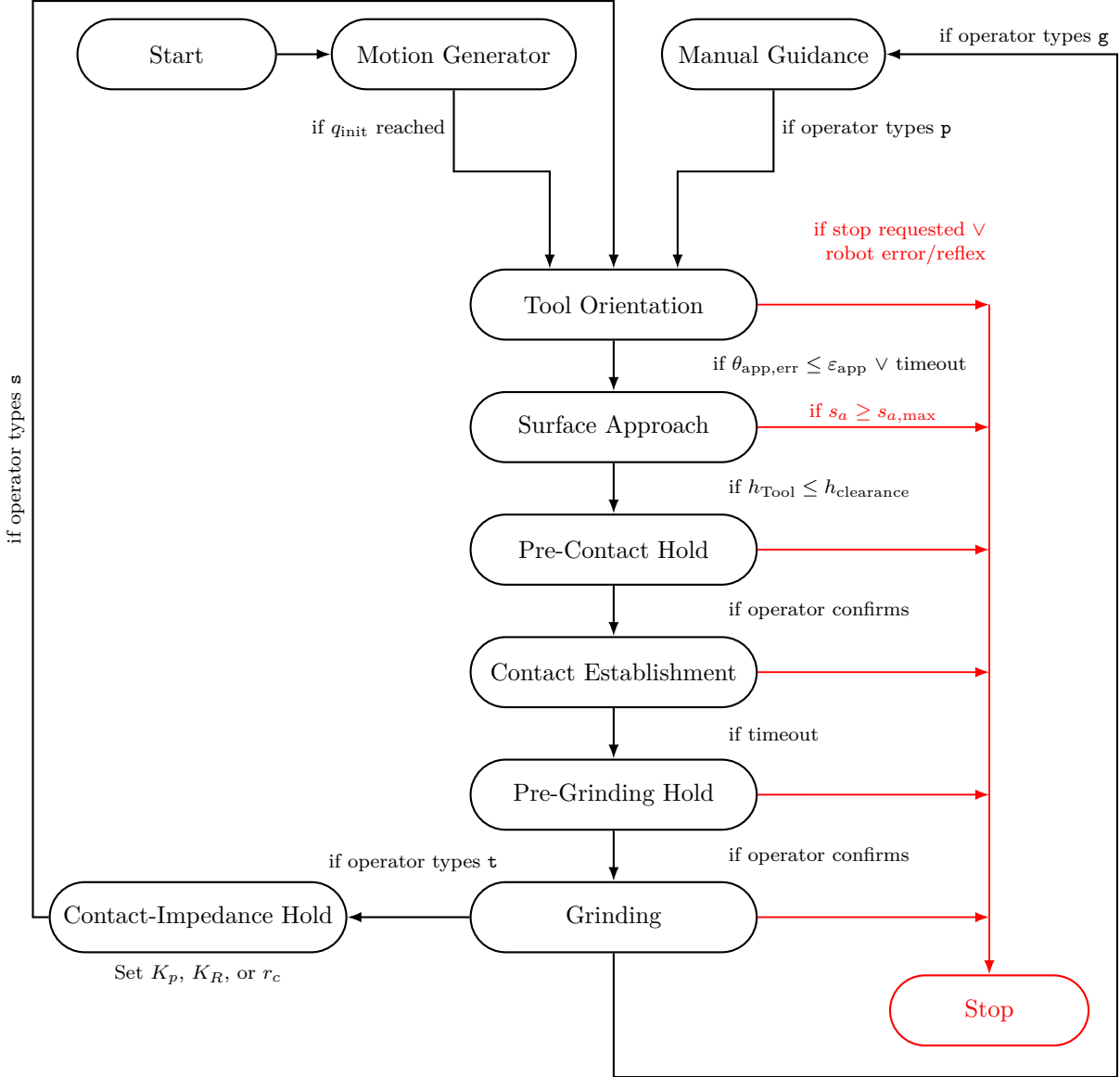


Figure 3.3: Surface-contact sequence and contact-impedance tuning loop.

3.2.1 Tool Orientation Reference and Tool Geometry

The surface-frame directions introduced in Section 2.5 are used to define the pre-contact tool orientation. The calibrated tool-face normal $n_{Tool,EE}$ is expressed in end-effector

coordinates. Its current direction in the robot base frame is

$$n_{\text{Tool}} = R_{\text{EE}} n_{\text{Tool,EE}}. \quad (3.1)$$

The configured orientation offset combines small rotations about the two surface tangents. Composing the two rotation matrices gives the exact combined orientation, which depends on the order in which the rotations are applied. For small configured angular offsets, the pre-contact orientation offset is represented to first order in the surface tangent plane by

$$\theta_{\text{offset}} \approx \theta_{\text{offset},t_1} t_1 + \theta_{\text{offset},t_2} t_2 \in \mathbb{R}^3. \quad (3.2)$$

Here t_1 and t_2 are the rotation axes, and each product is the first-order contribution of one rotation, expressed in radians. The direction of θ_{offset} defines the rotation axis of the combined offset, while its magnitude $\|\theta_{\text{offset}}\|$ defines its rotation angle. For a non-zero offset, the corresponding unit rotation axis is

$$u_{\text{offset}} = \frac{\theta_{\text{offset}}}{\|\theta_{\text{offset}}\|}, \quad (3.3)$$

and the corresponding rotation follows from the axis–angle relation of Equation (2.7),

$$R_{\text{offset}} = R(u_{\text{offset}}, \|\theta_{\text{offset}}\|).$$

This parameterisation gives a single axis–angle rotation, so the controller applies one rotation rather than a sequence of two.

A tool face parallel to the configured surface has the tool-normal direction $-n_s$. Applying the configured rotation gives the desired tool-normal direction

$$n_{\text{Tool},d} = R_{\text{offset}}(-n_s). \quad (3.4)$$

The controller determines the target end-effector orientation that brings the tool-normal direction to $n_{\text{Tool},d}$. An optional rotation about $n_{\text{Tool},d}$ sets the in-plane orientation of the rectangular tool face.

The rectangular tool face is represented in end-effector coordinates by its centre offset

$r_{\text{face,EE}}$, half-width $r_{\text{width,EE}}$, and half-length $r_{\text{length,EE}}$. These define the four corner offsets

$$r_{i,\text{EE}} \in \{r_{\text{face,EE}} \pm r_{\text{width,EE}} \pm r_{\text{length,EE}}\}, \quad i \in \{1, 2, 3, 4\}. \quad (3.5)$$

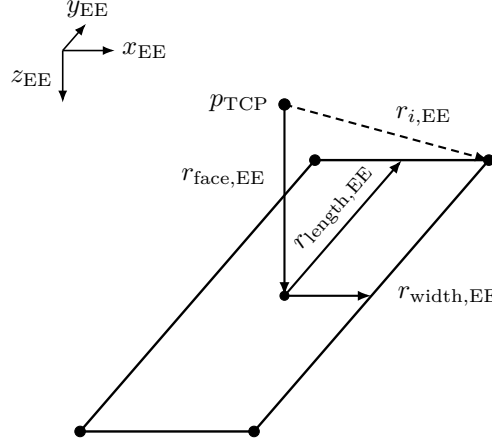


Figure 3.4: Calibrated tool face and its four corners.

Figure 3.4 shows these offsets in the end-effector frame.

The corner positions are used during Surface Approach to determine the clearance between the tool face and the configured surface.

3.2.2 Tool Orientation

Tool Orientation rotates the end effector from its orientation at state entry to the configured pre-contact orientation while maintaining the TCP position. The desired orientation is approached through a rate-limited interpolation.

After the configured minimum duration, the controller advances to Surface Approach when the enabled orientation errors satisfy their respective tolerances. The orientation timeout provides the alternative transition.

3.2.3 Surface Approach, Clearance and Tool-Point Selection

Surface Approach moves the TCP towards the configured surface along $-n_s$ while retaining the orientation reached during Tool Orientation. At the beginning of the state, denoted t_{start} , the controller captures the TCP position $p_{\text{TCP,start}}$. The commanded approach distance is

$$s_a(t) = \begin{cases} v_a(t - t_{\text{start}}), & t_{\text{start}} \leq t < t_{\text{end}}, \\ s_{a,\text{max}}, & t \geq t_{\text{end}}, \end{cases} \quad (3.6)$$

and the desired TCP position is

$$p_d(t) = p_{\text{TCP,start}} - s_a(t)n_s. \quad (3.7)$$

During the linear approach,

$$\dot{p}_d(t) = -v_a n_s.$$

After $s_{a,\text{max}}$ is reached, the desired position remains unchanged and $\dot{p}_d = 0$.

The TCP position alone does not describe the clearance of the rectangular tool face from the surface, particularly when the tool is inclined. The controller therefore calculates the position of each tool-face corner,

$$p_i(t) = p_{\text{TCP}}(t) + R_{\text{EE}}(t)r_{i,\text{EE}}, \quad i \in \{1, 2, 3, 4\}, \quad (3.8)$$

and its surface-normal clearance

$$h_i(t) = n_s^\top (p_i(t) - p_s). \quad (3.9)$$

The tool-face clearance is determined by the corner closest to the configured surface,

$$h_{\text{Tool}}(t) = \min_{i \in \{1, 2, 3, 4\}} h_i(t). \quad (3.10)$$

Surface Approach ends when the specified clearance is reached,

$$h_{\text{Tool}}(t) \leq h_{\text{clearance}}. \quad (3.11)$$

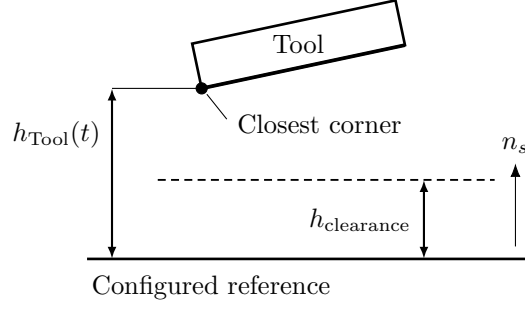


Figure 3.5: Tool-face clearance criterion.

Figure 3.5 illustrates this clearance condition.

At the clearance transition, the corner clearances also identify the part of the rectangular tool face closest to the configured surface. Corner i belongs to the selected set when

$$I_{\text{sel}} = \{i \in \{1, 2, 3, 4\} \mid h_i - h_{\text{Tool}} \leq \varepsilon_{\text{sel}}\}. \quad (3.12)$$

More than one corner can lie closest to the surface, and the tolerance ε_{sel} decides how close counts as equal. Exact equality does not occur in the measured geometry, so corners that are level within the calibration resolution have to be admitted by a bound.

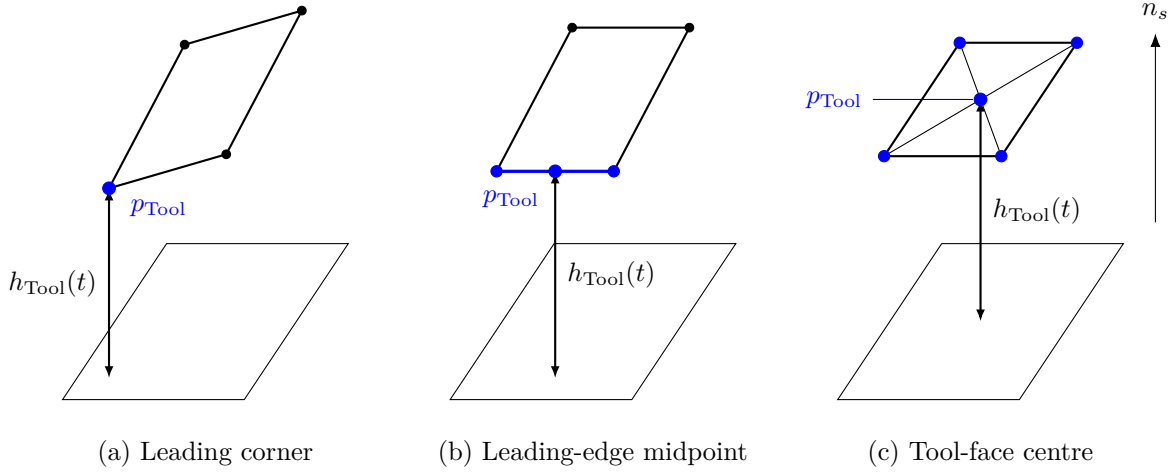


Figure 3.6: Principal selected tool points: (a) corner, (b) edge midpoint, and (c) tool-face centre.

Three principal cases follow for a rectangular face, drawn in Figure 3.6. A face inclined about both surface tangents meets the plane at one corner, and the set holds that corner alone. A face inclined about one tangent only has two corners of the leading edge equidistant from the plane, and both enter the set. A face parallel to the configured surface has all four corners equidistant, and the set holds all four. The tolerance is applied to each

corner separately, so a face that is near-level about both tangents can also admit three corners.

The controller takes the mean of the selected corner offsets. The three principal cases therefore give the leading corner, the midpoint of the leading edge, and the centre of the tool face. With

$$N = |I_{\text{sel}}|,$$

the corresponding selected tool offset is

$$r_{\text{Tool,EE}} = \frac{1}{N} \sum_{i \in I_{\text{sel}}} r_{i,\text{EE}}. \quad (3.13)$$

The vector $r_{\text{Tool,EE}}$ points from the TCP to the selected tool-face point and is expressed in end-effector coordinates. The selected tool offset is retained for the Contact Establishment trajectory. It is used to determine the initial tool point $p_{\text{Tool}}(t_{\text{start}})$ and its projection $p_{\text{Tool},0}$ onto the plane, from which the desired tool-point trajectory is generated up to the endpoint $p_{\text{Tool},d}(t_{\text{end}})$.

3.2.4 Pre-Contact Hold

The optional Pre-Contact Hold maintains the Cartesian pose reached at the end of Surface Approach. After operator confirmation, the controller captures the current measured pose and uses it to initialise Contact Establishment. With Pre-Contact Hold disabled, reaching the clearance condition triggers the direct transition from Surface Approach to Contact Establishment.

3.2.5 Contact Establishment

Let t_{start} denote the beginning of Contact Establishment. Using the selected tool offset $r_{\text{Tool,EE}}$ from Section 3.2.3, the corresponding tool point at this instant is

$$p_{\text{Tool}}(t_{\text{start}}) = p_{\text{TCP}}(t_{\text{start}}) + R_{\text{EE}}(t_{\text{start}})r_{\text{Tool,EE}}. \quad (3.14)$$

The controller projects this point along the surface normal onto the configured surface plane,

$$p_{\text{Tool},0} = p_{\text{Tool}}(t_{\text{start}}) - n_s n_s^\top (p_{\text{Tool}}(t_{\text{start}}) - p_s). \quad (3.15)$$

The projected point $p_{\text{Tool},0}$ defines the zero position of the contact-trajectory coordinate s_c . The initial value is

$$s_c(t_{\text{start}}) = -n_s^\top (p_{\text{Tool}}(t_{\text{start}}) - p_s). \quad (3.16)$$

This value places the desired tool point at its position at the beginning of Contact Establishment and provides a continuous position reference.

The contact-trajectory coordinate increases at the configured speed v_c until the endpoint $s_{c,\text{max}}$ is reached at t_{end} :

$$s_c(t) = \begin{cases} s_c(t_{\text{start}}) + v_c (t - t_{\text{start}}), & t_{\text{start}} \leq t < t_{\text{end}}, \\ s_{c,\text{max}}, & t \geq t_{\text{end}}. \end{cases} \quad (3.17)$$

The corresponding desired tool point is

$$p_{\text{Tool},d}(t) = p_{\text{Tool},0} - s_c(t) n_s. \quad (3.18)$$

Since $p_{\text{Tool},0}$ lies on the configured surface plane, s_c gives the signed distance of the desired tool point from this plane along the surface-normal direction. At $s_c = s_{c,\text{max}}$, the desired endpoint lies $s_{c,\text{max}}$ below the configured plane.

The desired orientation is held at the measured orientation at the beginning of Contact Establishment,

$$R_d(t) = R_{\text{EE}}(t_{\text{start}}). \quad (3.19)$$

The Cartesian impedance controller regulates the TCP. The selected tool offset therefore gives the corresponding desired TCP position as

$$p_d(t) = p_{\text{Tool},d}(t) - R_{\text{EE}}(t_{\text{start}})r_{\text{Tool,EE}}. \quad (3.20)$$

The reference construction can therefore be summarised as

$$r_{\text{Tool,EE}} \rightarrow p_{\text{Tool}}(t_{\text{start}}) \rightarrow p_{\text{Tool},0} \rightarrow p_{\text{Tool},d}(t) \rightarrow p_d(t)$$

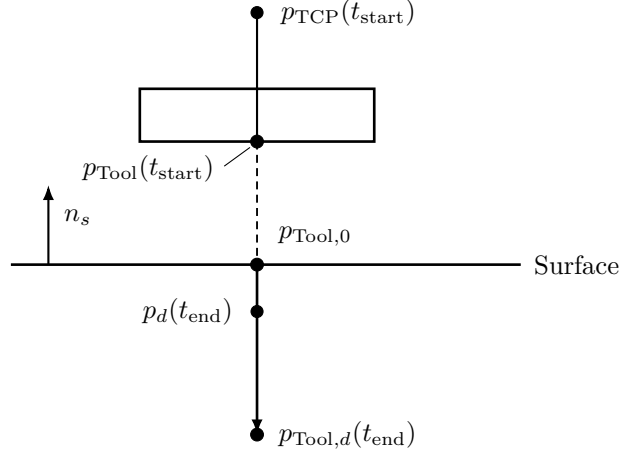


Figure 3.7: Contact-establishment reference generation.

Figure 3.7 draws this reference construction.

During the linear part of the contact trajectory,

$$\dot{p}_d = -v_c n_s,$$

and after $s_{c,\max}$ is reached,

$$\dot{p}_d = 0.$$

Contact limits the commanded translation towards the surface, and the resulting position error generates the pressing force through the Cartesian impedance law. The held orientation reference and finite rotational compliance allow the contact interaction to generate the rotational response.

During Contact Establishment, the Cartesian impedance is configured with higher translational stiffness along t_1 and t_2 and lower stiffness along n_s , providing tangential position retention together with compliant normal pressing. The rotational stiffness follows the

complementary directional choice: low stiffness about t_1 and t_2 permits contact-induced alignment, while higher stiffness about n_s maintains the in-plane tool orientation.

3.2.6 Pre-Grinding Hold

Pre-Grinding Hold begins when Contact Establishment reaches its independent timeout. The final Contact Establishment position and orientation references are retained, while the Cartesian impedance controller remains active. The controller therefore continues to generate the corresponding force and moment commands while waiting for operator confirmation.

During Pre-Grinding Hold, if the physical surface orientation is changed manually, the resulting contact moment rotates the tool towards the changed surface orientation. The low rotational stiffness generates a comparatively small restoring commanded moment towards the held desired orientation, allowing the contact-induced moment to dominate and produce passive alignment with the changed surface. With a displaced CoC, the additional commanded moment $r_c \times f$ biases the rotational response towards one direction. Depending on the selected CoC displacement, this moment can reinforce or counteract the contact-induced rotation.

After operator confirmation, the sequence proceeds to Grinding.

3.2.7 Grinding

After operator confirmation, the Grinding state retains the normal pressing reference established during Contact Establishment and superimposes a smooth tangential motion, with the impedance defined at the TCP.

The experimental evaluation in this thesis concerns the angular error during Contact Establishment. Grinding remains the implemented subsequent state of the complete surface-contact sequence.

3.3 Cartesian Pose Hold

Cartesian Pose Hold begins after the robot reaches the stored initial joint configuration q_{init} , or from the current guided configuration q when the operator types **h** in Manual Guidance. At state entry t_0 , the controller captures $p_d = p_{\text{EE}}(t_0)$ and $R_d = R_{\text{EE}}(t_0)$, and maintains these references with a dedicated hold impedance expressed in the base frame.

Four null-space modes can be selected without changing the Cartesian reference:

- Mode 0 applies no null-space torque.
- Mode 1 applies projected damping, with d_{null} adjustable.
- Mode 2 applies singular-value conditioning, with k_{σ} adjustable.
- Mode 3 applies both terms together, with both parameters adjustable.

Entering another number switches modes while the Cartesian reference and impedance remain unchanged.

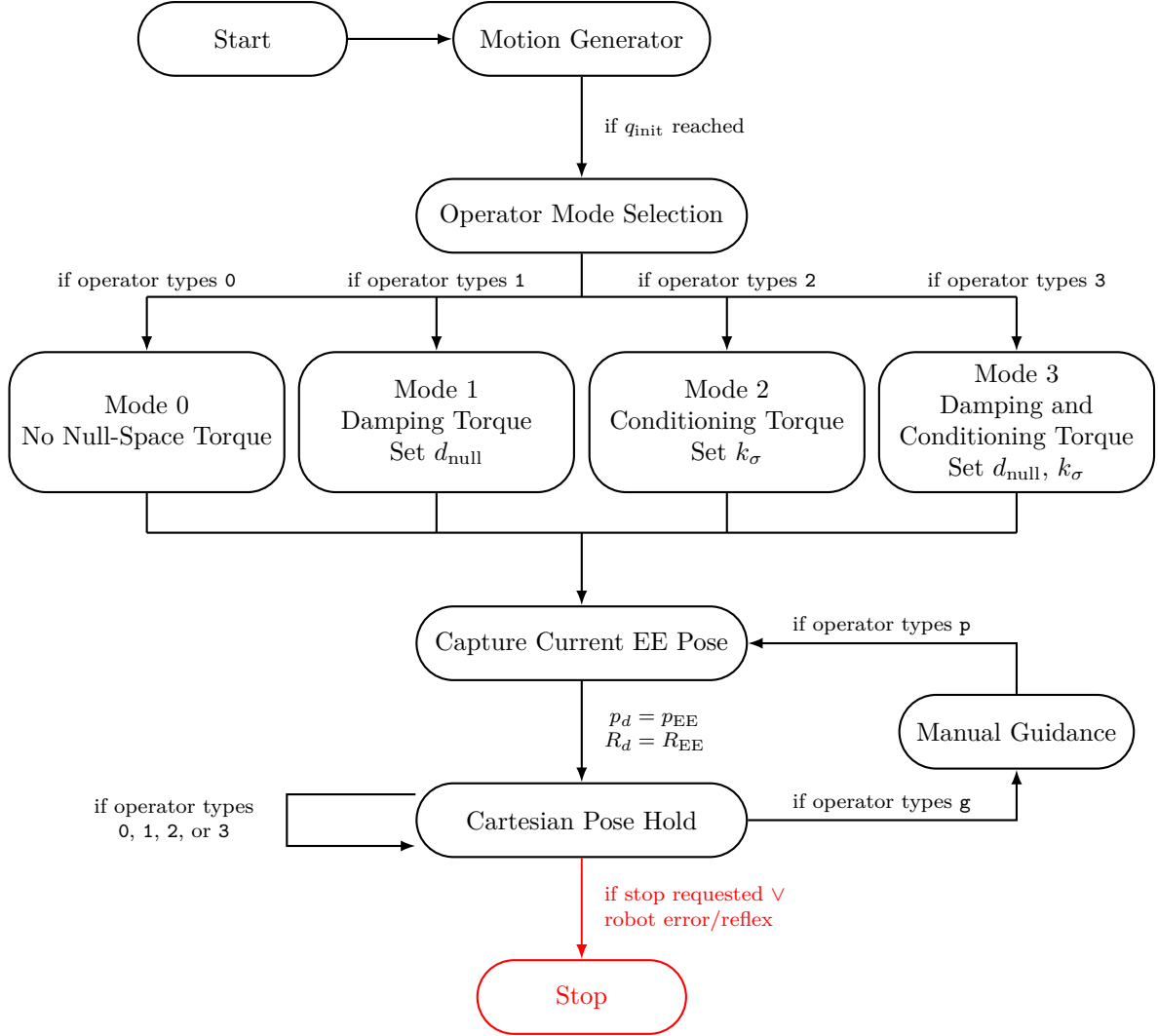


Figure 3.8: Operator-selectable null-space modes in Cartesian Pose Hold.

The implementation specifies two numerical choices beyond Section 2.8. The pseudoinverse threshold is the configured relative tolerance multiplied by the largest current singular value. The resulting joint-torque projector is symmetrised numerically, and the controller reuses that same projector to calculate the logged projected joint velocity $\dot{q}_{null}$. For singular-value conditioning, the controller evaluates the Jacobian at $q \pm \alpha_{probe} v_7$ using the current robot transformations, and compares the smallest singular value at each probe. The configured deadband sets the conditioning torque to zero.

3.4 Robot-Side Safety and Command Handling

The complete joint-torque command of Equation (2.73) is returned directly to libfranka. The configured robot-side collision and reflex monitoring provide the protection layer used

during the experiments. Its joint-torque and Cartesian-wrench thresholds are recorded in Section 4.1.3. A robot communication, motion-generation or torque-control error leaves the callback through `franka::Exception` and ends the control session.

A libfranka joint-motion generator brings the robot to the configured initial posture before torque control starts, unless the experiment starts from a pose captured after Manual Guidance. The posture used in the experiments is recorded in Appendix C. Operator input and file writing remain outside the real-time thread, and the torque callback operates only on prepared in-memory data.

3.5 Real-Time Data Recording

During torque control, the controller stores the experimental signals in a preallocated, fixed-capacity in-memory ring buffer. Each row carries the measured state, the active Cartesian reference, the calculated pose error and wrench, the controller state, the null-space configuration, and the returned joint torque. When the allocated capacity is reached, the write index wraps and new rows replace the oldest. After torque control ends, the remaining buffer contents are written to the CSV file in chronological order.

Chapter 4

Experimental Methodology

This chapter defines the experimental system, the calibration procedures, the surface-contact test matrix, and the evaluation quantities. The surface-contact experiments varied the configured orientation offset, the directional Cartesian stiffness, and the virtual CoC, while the remaining experimental conditions were held constant. The primary outcome was the end-effector rotation generated during Contact Establishment.

A separate Cartesian pose-hold experiment evaluated projected null-space damping and singular-value conditioning under a repeatable internally commanded disturbance, referred to below as the virtual disturbance.

4.1 Experimental System

4.1.1 Robot, Tool, and Surface

The experiments were performed with a seven-degree-of-freedom Franka Emika Panda operated through the Franka Control Interface. Joint-torque commands were transmitted through libfranka at the nominal control rate of 1 kHz, corresponding to a sampling period of 1 ms. The real-time controller, Cartesian impedance law, contact geometry, and state sequence are described in Chapter 3.

The rectangular grinding tool was held by the Franka Hand using custom gripper fingers. The tool face measured 40×120 mm, with the width along X_{EE} and the length along Y_{EE} . Its centre was located 20 mm along $+Z_{EE}$ from the end-effector origin. The tool was

clamped by custom pads screwed to the gripper fingers.

The plane plate was mounted on a raised base board, and every experiment used dry contact. Panel (a) of Figure 4.1 shows the physical arrangement. Panel (b) separates the configured surface reference, which is a controller quantity, from the physical plane of the set-up. Section 4.2 describes how the configured surface reference and the tool normal were established separately.

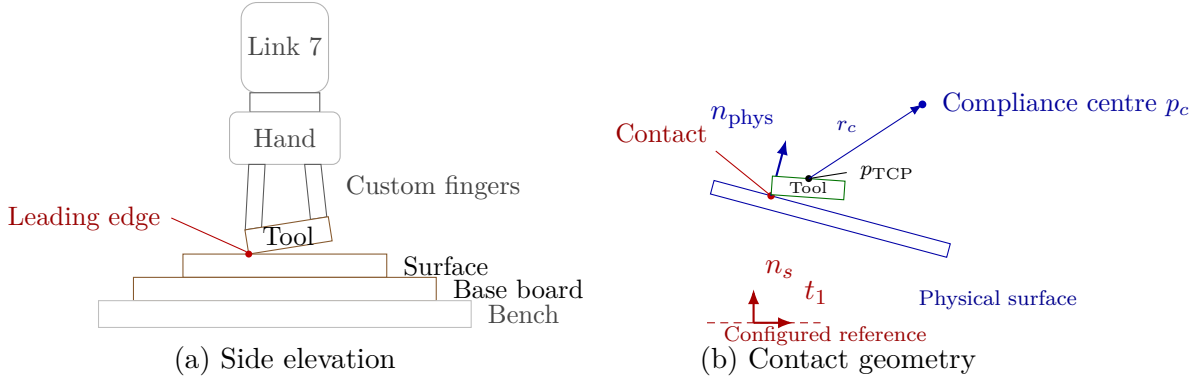


Figure 4.1: Experimental system: (a) physical set-up and (b) surface-frame contact geometry.

4.1.2 Real-Time Computer and Software

The controller computer ran Ubuntu 20.04.3 with the 5.9.1-rt20 Linux kernel and the PREEMPT_RT patch. The controller was compiled using g++ 9.3, C++14, and optimisation level -O2. Eigen 3.3.7 was used for matrix calculations and libfranka 0.7.1 for communication with the robot.

Before the measurements, the computer-robot connection was checked with the `communication_test` program supplied with libfranka 0.7.1 [6]. The program reports the fraction of control commands received by the robot, and issues a communication-quality warning when the mean command-success rate falls below 0.90. The experimental system passed this criterion, which confirms that the communication quality met the libfranka requirement for operation through the nominal 1 kHz interface.

4.1.3 Operating and Safety Configuration

Before torque control started, the acceleration and nominal joint-torque collision thresholds were both set to 140 N m for every joint. The Cartesian thresholds were set to 240 N for the force channels and 240 N m for the moment channels. These values configured the robot-side collision and reflex monitoring and were kept unchanged throughout the measurements.

The complete experiment-specific controller configuration is given in Appendix C.

4.2 Surface-Reference and Tool-Normal Calibration

To establish the configured surface reference and tool normal, two independent procedures were performed before the contact experiments. Surface-reference construction determined p_s , n_s , and R_{surface} in the robot base frame, while tool-normal calibration determined $n_{\text{Tool,EE}}$ in end-effector coordinates.

The configured surface normal n_s was obtained from a pose with the tool face seated on the physical plate. It supplies the calibrated surface reference used by the controller and the angular evaluation. Panel (b) of Figure 4.1 distinguishes this reference from the physical plate orientation.

4.2.1 Surface-Reference Construction

The complete tool face was seated on the physical plane, and one end-effector pose was recorded. At that pose the tool face lay against the plane, with $+Z_{\text{EE}}$ directed into it. The nominal $-Z_{\text{EE}}$ axis transformed into the base frame therefore defined the outward configured surface normal n_s . The transformed tool-face-centre offset defined the point p_s on the configured reference plane. The surface tangents and R_{surface} were then constructed as in Section 2.5. The resulting configured surface normal corresponded to rotations of -1.585° about the base-frame x -axis and 0.988° about the base-frame y -axis. Appendix C gives the complete numerical values.

4.2.2 Tool Normal Calibration

The complete tool face was seated on the plane at three yaw orientations. For each recorded pose, the physical tool-face normal corresponds to the same direction in the robot base frame. A unit vector in end-effector coordinates was therefore fitted so that its base-frame direction varied as little as possible across the three poses, which yields $n_{\text{Tool,EE}}$. The calibrated tool normal differed from the nominal $+Z_{\text{EE}}$ direction by approximately 1.56° . Appendix C gives the fitting relation and the numerical vector.

4.2.3 Calibrated Experimental Geometry

Table 4.1 lists the rectangular tool geometry and the tool-point selection tolerance used by the controller. Section 3.2.3 describes how these values determine the selected tool point, and Section 3.2.5 describes how the corresponding TCP reference is reconstructed.

Table 4.1: Calibrated tool geometry.

Quantity	Value in end-effector coordinates
Tool-face centre	$[0, 0, 0.020]^\top \text{ m}$
Half-width	$[0.020, 0, 0]^\top \text{ m}$
Half-length	$[0, 0.060, 0]^\top \text{ m}$
Tool-point selection tolerance, ε_{sel}	0.1 mm

4.3 Contact-Test Configuration and Procedure

Each surface-contact comparison varied one parameter. The calibrated geometry, the state sequence, the normal-press trajectory, the remaining stiffness and damping entries, and the null-space configuration were held constant.

4.3.1 Common Cartesian Stiffness and Damping

Before the compliance-centre transformation is applied, the state-dependent stiffness and damping matrices are diagonal in the configured surface frame $[t_1, t_2, n_s]$. Table 4.2 gives the entries used by every contact experiment, together with the damping factor ζ of the

corresponding state.

Table 4.2: Common Cartesian stiffness and damping factor by state.

State	$K_{p,S}$ [N/m]			$K_{R,S}$ [N m/rad]			ζ
	t_1	t_2	n_s	t_1	t_2	n_s	
Orientation and approach	150	150	150	180	180	90	1.9
Contact-establishment	2000	2000	350	5	5	50	1.0

The directional design these entries follow is stated in Section 3.2.5. The directional damping coefficients were calculated using the inertia-scaled formulation of Section 2.6.1.

4.3.2 Contact Sequence and Contact-Establishment Motion

The common trajectory and transition parameters are collected in Table 4.3. These settings were applied throughout the surface-contact experiments.

Table 4.3: Common state-transition and trajectory parameters.

Quantity	Value
Minimum orientation time	1.0 s
Tool-axis transition tolerance	0.016 rad
Spin transition tolerance	0.009 rad
Orientation timeout / maximum rate	5.0 s / 20°/s
Descent speed, v_a	0.1 m/s
Maximum descent distance, $s_{a,\max}$	0.640 m
Tool clearance, $h_{\text{clearance}}$	0.020 m
Minimum Contact Establishment time	2.0 s
Contact Establishment timeout	5.0 s
Press speed, v_c	0.080 m/s
Virtual press endpoint, $s_{c,\max}$	0.240 m
Hold before Contact Establishment	disabled
Hold before Grinding	enabled

Before each experiment, the active parameter set was loaded and the robot was moved to the configured initial posture. Surface Approach ended when the minimum clearance

of the rectangular tool face reached $h_{\text{clearance}}$. Contact Establishment then advanced the selected tool-point reference towards the surface, while holding the end-effector orientation captured at the clearance transition. Every contact experiment reached the configured 5s Contact Establishment timeout and then entered the operator-controlled hold before Grinding.

Throughout Cases A–D, projected null-space damping and singular-value conditioning remained active together in Mode 3, with $d_{\text{null}} = 2 \text{ N m s/rad}$ and $k_{\sigma} = 1 \text{ N m}$. These parameters were held constant throughout the surface-contact study and were therefore not experimental variables. The conditioning torque magnitude was kept at the comparatively low value of 1 N m to limit its influence on the angular behaviour during contact while keeping the null-space controller active.

4.3.3 Repetitions and Data Recording

The recorded signals and CSV structure are documented in Section 3.5 and Appendix B. Each parameter setting was evaluated in three repeated trials. The reported values are arithmetic means with sample standard deviations. Cases A–D comprised 31 parameter settings and 93 surface-contact trials. The 12 Cartesian pose-hold trials of Section 4.7 bring the complete experimental data set reported in this thesis to 105 trials.

4.4 Main Surface-Contact Experimental Matrix

The main surface-contact study separated the effects of the principal controller parameters for configured orientation offsets about t_1 . Case A established the TCP-centred baseline $r_c = 0$, for the nominal zero offset and for the positive- and negative-offset conditions about t_1 . Case B varied the rotational stiffness K_{R,t_1} , and Case C the translational stiffness along t_2 , K_{p,t_2} , in each instance with the other entry held constant. End-effector rotation about t_1 can be accompanied by tangential motion along t_2 . The translational stiffness K_{p,t_2} was therefore selected to examine its influence on the angular error. Case D varied the tangential compliance-centre position r_{c,t_2} for both directions of the measured angular offset θ_{meas,t_1} . Table 4.4 summarises the parameter variations. The TCP-centred reference points used in Cases B–D reuse the corresponding Case-A baseline

measurements. Table 4.4 therefore lists only the additional parameter settings evaluated in Cases B–D.

All non-zero compliance-centre displacements were configured in end-effector coordinates and therefore remained constant relative to the tool. The r_{c,t_2} values reported for Case D are the surface-frame projections of those tool-fixed displacements in the flat target orientation. The positions ± 90 and ± 100 mm were evaluated in a separate session with the same saved calibration and impedance parameters.

Table 4.4: Main surface-contact cases and parameter variations.

Case	Comparison parameter	Measured angular offset	Tested values	Trials
A	CoC at the TCP	0.69° , 9.31° and -9.41° about t_1	$r_c = 0$.	9
B	Rotational stiffness	9.31° to 9.32° about t_1	$K_{R,t_1} = 15$ and 50 N m/rad.	6
C	Translational stiffness	9.24° to 9.31° about t_1	$K_{p,t_2} = 300$ and 800 N/m.	6
D	Tangential CoC position	9.33° and -9.38° about t_1	$r_{c,t_2} = -100, -90, -80, -40, -20, -10, 10, 20, 40, 80, 90,$ and 100 mm.	72

4.5 Measured Angular Offset and Angular Error

The configured angular offset $\theta_{\text{offset},t_1}$ sets the pre-contact orientation about t_1 . At zero configured offset, the commanded tool face is parallel to the configured surface reference. The orientation reached can differ from this command. The experimental comparisons therefore use the measured pose-based angular offset θ_{meas,t_1} at contact entry.

The angular error $\theta_{\text{err},t_1}(t)$ describes the tool-normal orientation relative to the calibrated surface reference throughout Contact Establishment. Both angular quantities use the inward configured surface normal $-n_s$ as their reference and are resolved about t_1 . The measured angular offset is the value of this error at contact entry.

Figure 4.2 shows the entry offset and angular error for a schematic rotation about t_1 . The solid green and dashed blue arrows show the calibrated tool normals at contact entry and contact end. They are calculated from measured end-effector orientation. Both angle arcs run from the red inward calibrated surface normal $-n_s$ to the respective tool normal. The common origin compares orientations. The view places t_1 out of the page, with positive rotation anticlockwise by the right-hand rule.

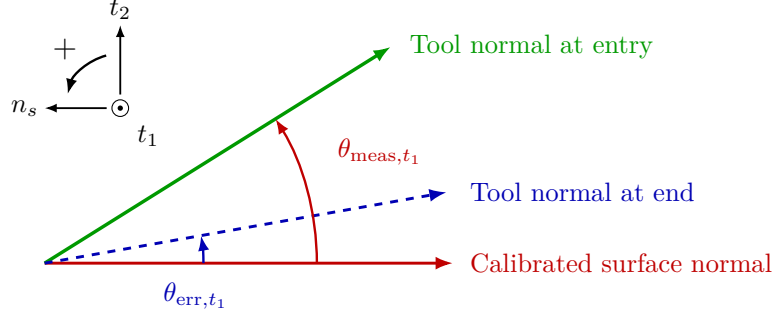


Figure 4.2: Measured angular offset and angular error.

4.5.1 Measured Pose-Based Angular Offset

At the start of Contact Establishment, t_{start} , the measured end-effector orientation gives the calibrated tool-normal direction

$$n_{\text{Tool}}(t_{\text{start}}) = R_{\text{EE}}(t_{\text{start}})n_{\text{Tool,EE}}. \quad (4.1)$$

The measured angular offset is represented by the rotation vector θ_{meas} . It describes the shortest rotation from the inward configured surface normal $-n_s$ to the entry tool-normal direction. For $\theta_{\text{meas}} \neq \mathbf{0}$, this rotation satisfies

$$R\left(\frac{\theta_{\text{meas}}}{\|\theta_{\text{meas}}\|}, \|\theta_{\text{meas}}\|\right)(-n_s) = n_{\text{Tool}}(t_{\text{start}}). \quad (4.2)$$

When the two directions coincide, $\theta_{\text{meas}} = \mathbf{0}$. Its component about t_1 is

$$\theta_{\text{meas},t_1} = t_1^\top \theta_{\text{meas}}. \quad (4.3)$$

The component θ_{meas,t_1} identifies the angular condition in every contact comparison, including the nominal zero-offset condition. The direction n_s is fixed throughout each trial.

4.5.2 Angular Error During Contact

The angular error was calculated throughout Contact Establishment from the measured end-effector orientation and the calibrated tool normal. At time t , the tool-normal direction in the base frame is

$$n_{\text{Tool}}(t) = R_{\text{EE}}(t)n_{\text{Tool,EE}}. \quad (4.4)$$

The angular-error vector $\theta_{\text{err}}(t)$ represents the shortest rotation from $-n_s$ to $n_{\text{Tool}}(t)$. For $\theta_{\text{err}}(t) \neq \mathbf{0}$, it satisfies

$$R\left(\frac{\theta_{\text{err}}(t)}{\|\theta_{\text{err}}(t)\|}, \|\theta_{\text{err}}(t)\|\right)(-n_s) = n_{\text{Tool}}(t). \quad (4.5)$$

The vector is zero when the normal directions coincide. The component about the first surface tangent is

$$\theta_{\text{err},t_1}(t) = t_1^\top \theta_{\text{err}}(t), \quad \theta_{\text{err},t_1}(t_{\text{start}}) = \theta_{\text{meas},t_1}. \quad (4.6)$$

Positive and negative values follow the right-hand rule about t_1 . A smaller magnitude indicates a smaller angular-error component relative to the calibrated surface. A change through zero indicates that the component has crossed the reference orientation. The complete normal-direction error also contains the component about the other surface tangent.

The controller evaluated the normal-to-normal rotation from the measured end-effector orientation during each control cycle. Its recorded vector points from the tool normal towards $-n_s$. The evaluation reverses this vector to obtain $\theta_{\text{err}}(t)$ with the same direction convention as the measured angular offset. Appendix B identifies the recorded fields. The comparison values were extracted from the controller's per-trial reports at the end of Contact Establishment, t_{end} . Those endpoint components were stored to 0.01° . The arithmetic means and sample standard deviations were calculated from these trial-level values. The time histories use the recorded samples from the complete contact interval of each illustrated trial.

The rotational spring continues to use the controller orientation error e_R relative to the

held entry reference. At contact entry, $e_R = \mathbf{0}$ while the surface-relative angular error can be non-zero.

4.6 Cartesian Wrench Evaluation

Two evaluations of the Cartesian wrench were carried out beside the main experimental matrix. Both are supplementary to Cases A–D and add no settings to that matrix. The first compares the commanded wrench with the model-estimated external wrench during contact establishment. The second applies a defined translational perturbation and a defined rotational perturbation, and compares the commanded wrench with the quasi-static impedance prediction.

Both evaluations use the model-estimated external wrench supplied by libfranka. It is obtained from the robot’s dynamic model rather than from a force/torque sensor, which is what the est index records. The estimator returns the spatial wrench acting on the stiffness frame $\{K\}$, expressed in the base frame, so its base-frame moment carries the lever from the base origin to that frame. In the experimental configuration ${}^{\text{EE}}T_K = I$, so $\{K\}$ coincides with the TCP and the moment is expressed about the TCP by subtracting that lever,

$$m_{\text{est},\text{TCP}} = m_{\text{est}} - p_{\text{TCP}} \times f_{\text{est}}.$$

The evaluated components are the normal force and the moment about the first surface tangent,

$$F_{n,\text{est}} = n_s^\top f_{\text{est}}, \quad M_{t_1,\text{est}} = t_1^\top m_{\text{est},\text{TCP}}.$$

The commanded components $F_n = n_s^\top f$ and $M_{t_1} = t_1^\top m$ are resolved on the same configured surface axes, so command and estimate are compared at one reference point and in one frame.

4.6.1 Commanded–Estimated Wrench Comparison

The first evaluation used one trial of the TCP-centred positive-offset condition, taken from the main contact matrix and evaluated without changing any setting. Both wrenches were resolved over the complete 5 s of Contact Establishment. The comparison is read over the stationary interval from 4 to 5 s, in which the press has settled and the references are constant. Each quantity is reported as the mean over that interval, and the estimate is compared with the command as a percentage of the commanded mean.

4.6.2 Quasi-Static Plausibility Assessment

The second evaluation tested the implemented impedance relation directly, against the stiffness values it was configured with. It ran in Contact-Impedance Hold with the compliance centre at the TCP, so $r_c = 0$ and no point-shift coupling acted, with the null-space torque inactive and the configured damping in use. The evaluated stiffness entries were $K_{p,n} = 1000 \text{ N/m}$ and $K_{R,t_1} = 15 \text{ N m/rad}$.

Two perturbations were applied while the controller held its captured Cartesian reference. The translational evaluation displaced the tool along the surface normal and held the displaced position. The rotational evaluation rotated the tool about t_1 while the normal load remained applied. Both perturbations were applied manually, so their achieved displacement and rotation were measured from the recorded pose.

The translational and rotational evaluations are treated separately. For each evaluation, the stationary condition immediately before the perturbation defines the reference state. The translational evaluation therefore uses the unloaded state as its reference, whereas the rotational evaluation uses the loaded state before rotation. This removes the moment already present under the normal load from the rotational comparison.

The quasi-static predictions follow from the configured stiffness and the change in measured pose relative to the corresponding reference state,

$$F_{n,\text{qs}} = K_{p,n} n_s^\top (e_p - e_{p,0}), \quad M_{t_1,\text{qs}} = K_{R,t_1} t_1^\top (e_R - e_{R,0}),$$

where $e_{p,0}$ and $e_{R,0}$ denote the pose-error values at the respective reference states. Time is reset to $t = 0$ at the beginning of each evaluation, and the reported values are obtained

from the stationary part of the corresponding response.

4.7 Secondary Null-Space Pose-Hold Experiment

Projected null-space damping and singular-value conditioning were active together during the surface-contact experiments. A separate Cartesian pose-hold experiment therefore evaluated the terms individually and together under the same repeatable virtual disturbance.

4.7.1 Experimental Configuration

The initial TCP pose was captured as the hold reference. The controller used the stiffness and damping of Table 4.5. The translational and rotational matrices were isotropic, so each matrix is the stated scalar value multiplied by the 3×3 identity matrix. All twelve trials in this comparison used fixed Cartesian damping.

Table 4.5: Isotropic Cartesian stiffness and damping of the pose-hold trials.

Quantity		Value
Translational stiffness	$K_{p,\text{hold}}$	2000 N/m
Rotational stiffness	$K_{R,\text{hold}}$	50 N m/rad
Translational damping	$D_{p,\text{hold}}$	50 N s/m
Rotational damping	$D_{R,\text{hold}}$	8 N m s/rad

The comparison comprised four null-space settings, with three trials per setting: no null-space torque, projected damping alone, singular-value conditioning alone, and both terms together. The enabled values were $d_{\text{null}} = 2 \text{ N m s/rad}$ and $k_{\sigma} = 2 \text{ N m}$. The probe step, singular-value deadband, and relative SVD threshold remained at 0.08 rad , 10^{-6} , and 10^{-4} , respectively. The combined-control trials were acquired in a later session with the same controller revision and matching saved experimental settings.

4.7.2 Commanded Disturbance

The disturbance was implemented as the joint-torque equivalent of a virtual point force applied at a selected location on link 3. In the link frame, the point was

$$r_p = \begin{bmatrix} 0 \\ 0 \\ 0.200 \end{bmatrix} \text{ m.} \quad (4.7)$$

At the initial posture q_{init} , the structural null direction $v_7(q_{\text{init}})$ is a joint-motion direction that preserves the end-effector pose to first order, and the translational point Jacobian $J(q_{\text{init}})$ maps it to the corresponding velocity of the selected link point. The fixed base-frame force direction was chosen along that velocity,

$$u_f = \frac{J(q_{\text{init}})v_7(q_{\text{init}})}{\|J(q_{\text{init}})v_7(q_{\text{init}})\|_2}. \quad (4.8)$$

This choice aligns the virtual force with the motion of the selected link point produced by the initial structural null direction, so the disturbance excites the redundant configuration while the pose-hold impedance opposes its Cartesian component.

The commanded force magnitude was $F_{\text{dist}} = 20 \text{ N}$. The virtual force and its equivalent joint torque were

$$f_{\text{dist}}(t) = F_{\text{dist}} s_{\text{dist}}(t) u_f, \quad \tau_{\text{dist}}(t) = J(q(t))^{\top} f_{\text{dist}}(t). \quad (4.9)$$

Here u_f is the unit force direction defined in Equation (4.8), and $s_{\text{dist}}(t) \in [0, 1]$ is a dimensionless time-scaling function that switches the commanded virtual disturbance smoothly on and off. It was defined as

$$s_{\text{dist}}(t) = \begin{cases} 0, & t < 5 \text{ s}, \\ \frac{1}{2} \left[1 - \cos\left(\pi \frac{t - 5 \text{ s}}{2 \text{ s}}\right) \right], & 5 \text{ s} \leq t < 7 \text{ s}, \\ 1, & 7 \text{ s} \leq t < 8 \text{ s}, \\ \frac{1}{2} \left[1 + \cos\left(\pi \frac{t - 8 \text{ s}}{1 \text{ s}}\right) \right], & 8 \text{ s} \leq t < 9 \text{ s}, \\ 0, & t \geq 9 \text{ s}. \end{cases} \quad (4.10)$$

The disturbance increased smoothly from zero to the full commanded magnitude between 5 and 7 s, remained at full magnitude until 8 s, and decreased smoothly to zero by 9 s. The equivalent disturbance torque was bounded by

$$\|\tau_{\text{dist}}\|_2 \leq 3 \text{ N m}.$$

The controller generates a repeatable joint-torque excitation by mapping the commanded point force through $J^\top$. Its Cartesian component is opposed by the pose-hold impedance, and Equation (4.11) evaluates the resulting position retention.

In every trial the selected null-space law was active from the start of the pose-hold state. The first 5 s formed a pre-disturbance settling interval, during which the conditioning torque of the singular-value-conditioning settings could already move the redundant configuration. The disturbance then acted from 5 to 9 s, followed by an observation interval until the end of the trial. Each selected secondary term remained active from state entry through the complete trial. The singular-value-conditioning torque acted before, during, and after the disturbance interval.

4.7.3 Evaluation Quantities

All quantities in this subsection are evaluated over the interval in which the virtual disturbance acts, $t \in [5, 9] \text{ s}$. For the time-history plots, elapsed time from disturbance onset is $t_d = t - 5 \text{ s}$, so the evaluated interval is displayed from 0 to 4 s.

The null-space controller is intended to modify the redundant joint configuration while preserving the Cartesian pose. The Cartesian position error evaluates TCP position

retention during the disturbance. A deviation below 2 mm is treated as acceptable position retention, so a trial satisfies the criterion when

$$\max_{5s \leq t \leq 9s} \|e_p(t)\|_2 < 2 \text{ mm}, \quad (4.11)$$

where $e_p(t)$ is the Cartesian position-error vector.

The redundant joint motion is evaluated from the projected joint velocity defined in Equation (2.64). The vector $\dot{q}_{\text{null}} \in \mathbb{R}^7$ has units of rad/s and represents the component of the measured joint velocity that lies in the instantaneous kinematic null space.

The cumulative joint motion is

$$E_N = \int_{5s}^{9s} \|\dot{q}_{\text{null}}(t)\|_2 dt. \quad (4.12)$$

Taking the norm before integration makes motion in opposite directions accumulate. The cumulative projected joint motion E_N evaluates motion across all seven joints and is particularly relevant to the damping comparison. Its unit is radians, and the values are expressed in degrees. The cumulative time history evaluates the same integral from disturbance onset to each displayed instant.

The conditioning indicator $\sigma_{\min}(t)$ was evaluated throughout the disturbance interval. Its time history shows how the local Jacobian conditioning changes while the disturbance acts. Each trial was also characterised by its lowest value over the interval,

$$\min_{5s \leq t \leq 9s} \sigma_{\min}(t).$$

The minimum is calculated for each trial before the three-trial mean and sample standard deviation are formed.

The measured joint angle q_1 was also evaluated to compare the joint-angle histories. Joint 1 received the largest peak absolute component of the equivalent disturbance torque τ_{dist} in each of the twelve selected trials. Its angle was therefore used to illustrate the temporal response, while E_N provides the overall motion measure across all seven joints. The angle was sampled at a nominal 20 Hz, using the measured timestamps. The joint-angle change

Δq_1 was calculated relative to disturbance onset:

$$\Delta q_1(t_d) = q_1(t_d + 5 \text{ s}) - q_1(5 \text{ s}).$$

The values are expressed in degrees. Each plotted curve is the mean of three trials at common measured timestamps. Shaded bands show one sample standard deviation. No interpolation or smoothing was applied. Subtracting each trial's onset angle preserves positive and negative motion in Δq_1 . A joint can move away from its onset angle and later return close to it while E_N remains positive. A near-zero final Δq_1 establishes only that joint 1 returned near its onset angle; it does not establish that every joint returned or that little cumulative motion occurred. Both panels cover the full disturbance interval.

Chapter 5

Results and Discussion

The surface-contact experiments measured how the controller parameters affected the angular error during Contact Establishment. The primary quantity is the angular-error component $\theta_{\text{err},t_1}(t_{\text{end}})$ relative to the calibrated surface reference, defined in Section 4.5. Its magnitude gives the remaining error about the first surface tangent. Positive and negative values identify the rotational direction relative to that reference. A change through zero reverses the first-tangent error direction.

The measured pose-based angular offset θ_{meas,t_1} is the value at contact entry and identifies every angular condition. At the TCP-centred condition, the nominal-zero command gave $\theta_{\text{meas},t_1} = 0.69^\circ$. The 10° and -10° commands produced mean measured offsets of 9.31° and -9.41° , respectively. These are named the positive-offset and negative-offset conditions throughout this chapter.

Each contact parameter setting was repeated three times. The comparison values are arithmetic means of the angular errors from those repetitions. Appendix D tabulates the corresponding sample standard deviations. The endpoint components were stored to 0.01° , as described in Section 4.5.2.

5.1 Controller Wrench Evaluation

Both evaluations follow the procedure of Section 4.6, which defines the model-estimated components and the stationary intervals they are read over. Each figure below shades the interval its values are taken from. Their results are supplementary to Cases A–D.

5.1.1 Commanded and Model-Estimated Wrench Comparison

Figure 5.1 compares the commanded and model-estimated Cartesian wrench during one TCP-centred Contact Establishment trial. In the stationary interval, the model-estimated normal force differed from the commanded value by 2.32 %, while the model-estimated moment about the first surface tangent differed by 4.87 %. The commanded and estimated quantities therefore showed close agreement in sign and magnitude during the stationary contact condition. Panel (a) carries the normal force and panel (b) the moment about t_1 .

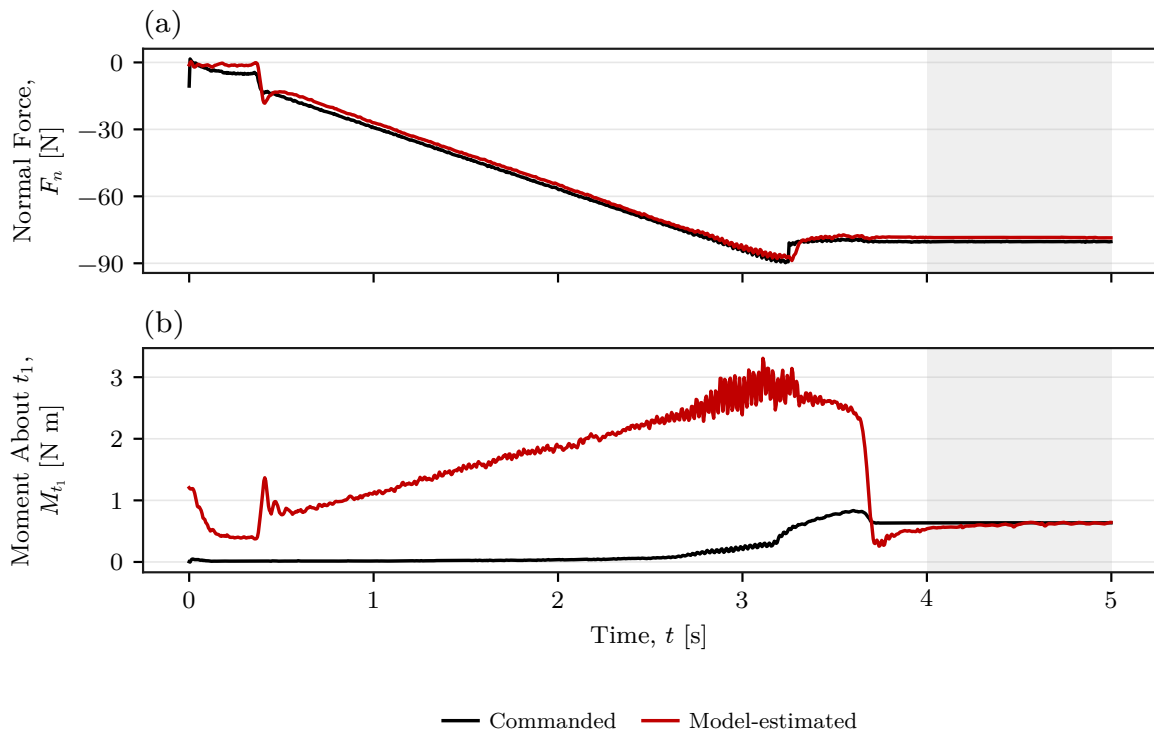


Figure 5.1: Commanded and model-estimated wrench during Contact Establishment.

5.1.2 Quasi-Static Plausibility Assessment

The translational perturbation reached -19.65 mm relative to its reference state, while the rotational perturbation reached 6.565° about t_1 relative to its reference state. The quasi-static predictions are evaluated at these measured displacements using $K_{p,n} = 1000$ N/m and $K_{R,t_1} = 15$ N m/rad.

The quasi-static prediction of the normal force is -19.650 N, against a commanded mean of -19.640 N and a model-estimated mean of -22.292 N. In the stationary interval, the quasi-static spring prediction differed from the commanded force by 0.05 %, while

the model-estimated force differed from the command by 13.50 %. The close agreement between the spring prediction and the commanded value supports the plausibility of the implemented translational impedance relation.

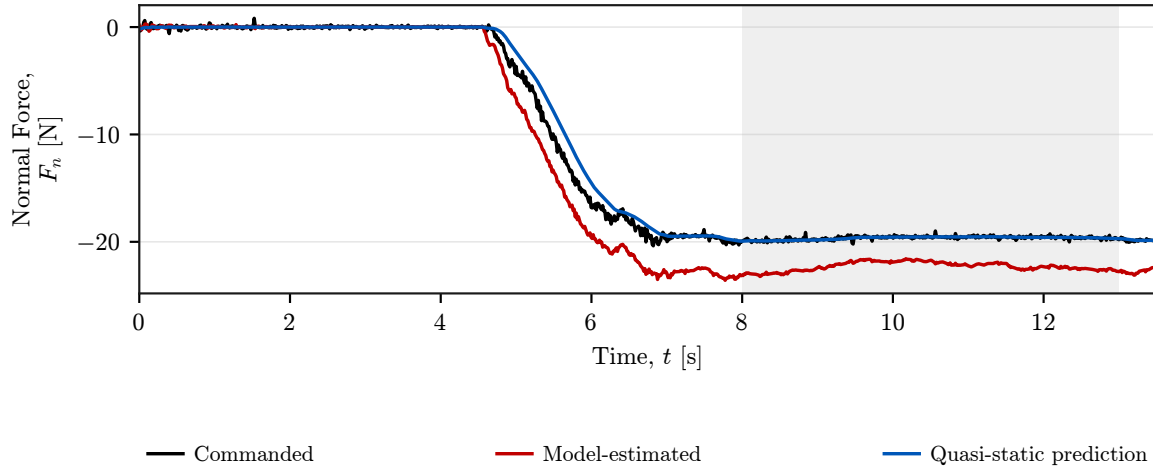


Figure 5.2: Quasi-static normal-force plausibility assessment.

For the rotation, the quasi-static prediction and the commanded mean are both 1.719 N m, and the model-estimated mean is 1.852 N m. The quasi-static spring prediction and the commanded moment agreed to the reported precision, while the model-estimated moment differed from the command by 7.75 %. This comparison supports the plausibility of the implemented rotational impedance relation.

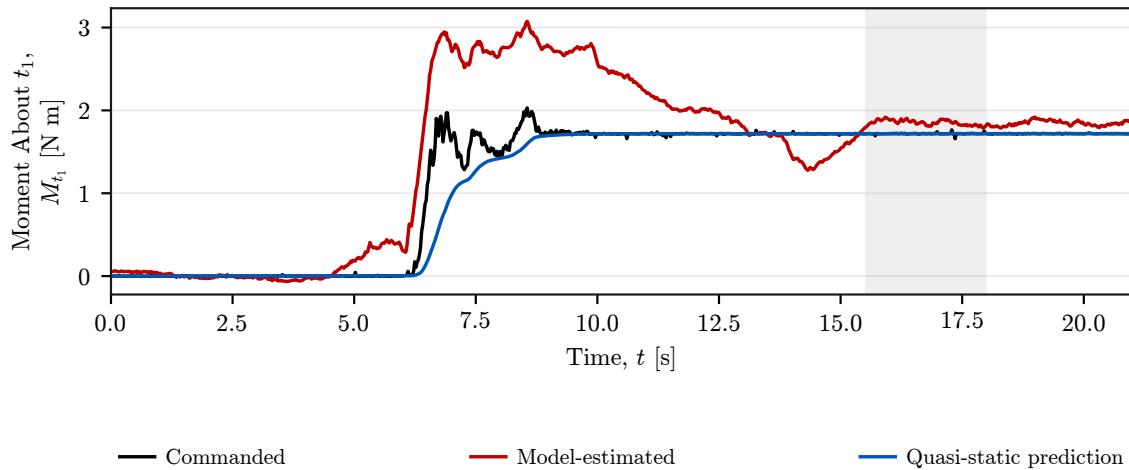


Figure 5.3: Quasi-static moment plausibility assessment about the first surface tangent.

5.2 Main Surface-Contact Results

5.2.1 Baseline Angular Error with the CoC at the TCP: Case A

In Case A, the CoC was placed at the TCP, so $r_c = 0$ and the CoC-induced moment $r_c \times f$ was zero. Contact Establishment was tested with three configured pre-contact orientation offsets about t_1 : 0° , 10° , and -10° . These conditions establish the TCP-centred baseline for the following stiffness and CoC-position comparisons.

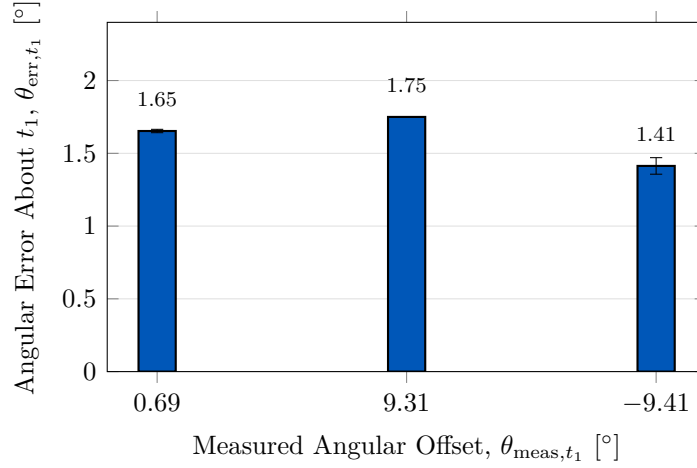


Figure 5.4: Angular error with the CoC at the TCP.

For the positive-offset condition, the angular error decreased from the entry value of 9.31° to 1.75° at contact end in Figure 5.4. For the negative-offset condition, it changed from -9.41° to 1.41° . The negative-offset component crossed zero and ended with a positive value. The angular error magnitudes were smaller than their respective entry magnitudes by 81.2% and 85.0%.

For the nominal-zero condition, the angular error increased from 0.69° to 1.65° . This condition therefore showed an increase in the first-tangent error. The TCP-centred condition reduced that error for both larger entry offsets, with different final values and a zero crossing in the negative-offset condition.

5.2.2 Rotational Stiffness: Case B

Case B investigated how the rotational stiffness about t_1 affects the angular error. Contact Establishment was performed with the positive 10° configured orientation offset while

K_{R,t_1} was compared at 5, 15, and 50 N m/rad. The 5 N m/rad condition is the TCP-centred baseline from Case A, while the remaining controller settings were kept unchanged.

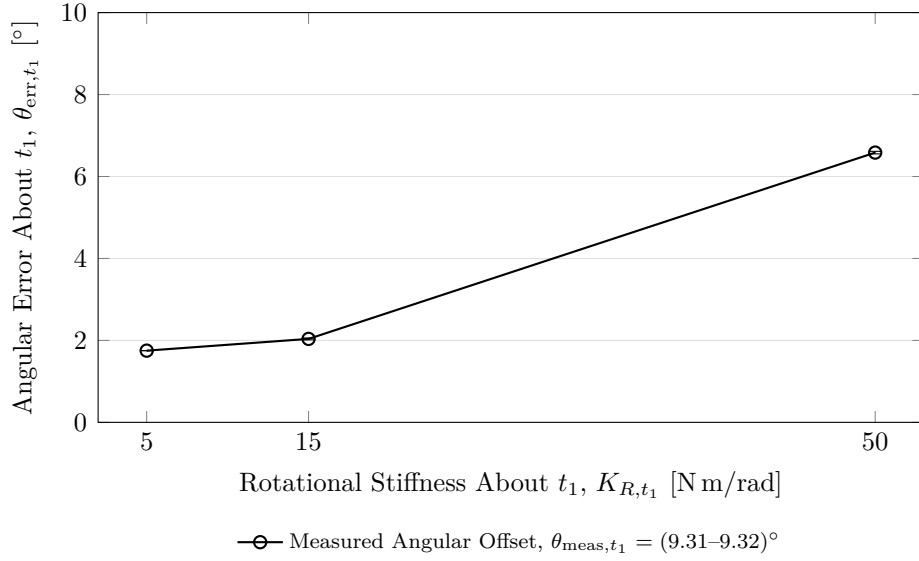


Figure 5.5: Angular error by rotational stiffness.

Increasing K_{R,t_1} from 5 to 50 N m/rad increased the angular error from 1.75° to 6.58° in Figure 5.5. The intermediate stiffness of 15 N m/rad gave 2.04° . The endpoint increase across the tested range was 4.83° , or 276.2 % of the error at 5 N m/rad. Higher rotational stiffness resisted the contact-induced end-effector rotation and left a larger angular error relative to the calibrated surface.

5.2.3 Translational Stiffness Along t_2 : Case C

Case C investigated the influence of the translational stiffness along t_2 , perpendicular to the measured rotation about t_1 . Contact Establishment was performed with the positive 10° configured orientation offset while K_{p,t_2} was compared at 300, 800, and 2000 N/m. The rotational stiffness was kept at $K_{R,t_1} = 5$ N m/rad, while the translational stiffness along t_1 remained $K_{p,t_1} = 2000$ N/m. The $K_{p,t_2} = 2000$ N/m condition is the corresponding TCP-centred reference from Case A.

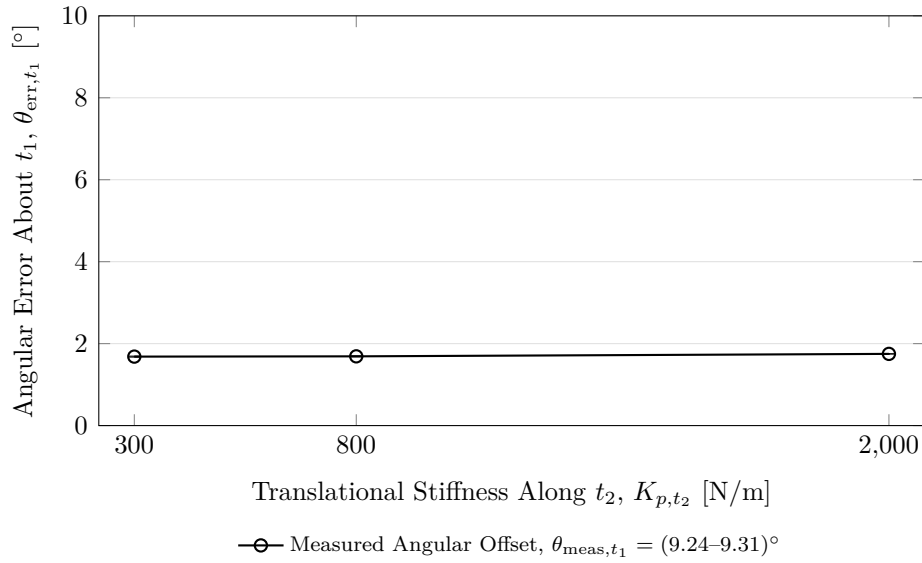


Figure 5.6: Angular error by translational stiffness along t_2 .

The angular errors were 1.68° , 1.69° and 1.75° at translational stiffnesses of 300, 800 and 2000 N/m, respectively, in Figure 5.6. Their span was 0.07° , or 3.8 % of the TCP-centred reference error at 2000 N/m. This variation was considerably smaller than the rotational-stiffness effect. The result concerns translation along t_2 , with the t_1 translational stiffness held constant.

5.2.4 Effect of Tangential CoC Position: Case D

Case D investigated how the tangential CoC position affects the angular error. The CoC was displaced along t_2 , perpendicular to the measured rotation about t_1 , and Contact Establishment was performed for both the positive 10° and negative -10° configured orientation offsets. The compared CoC positions were $r_{c,t_2} = -100, -90, -80, -40, -20, -10, 0, 10, 20, 40, 80, 90$, and 100 mm, where $r_{c,t_2} = 0$ is the TCP-centred reference from Case A. Cartesian stiffness, damping, and the remaining controller settings were kept unchanged.

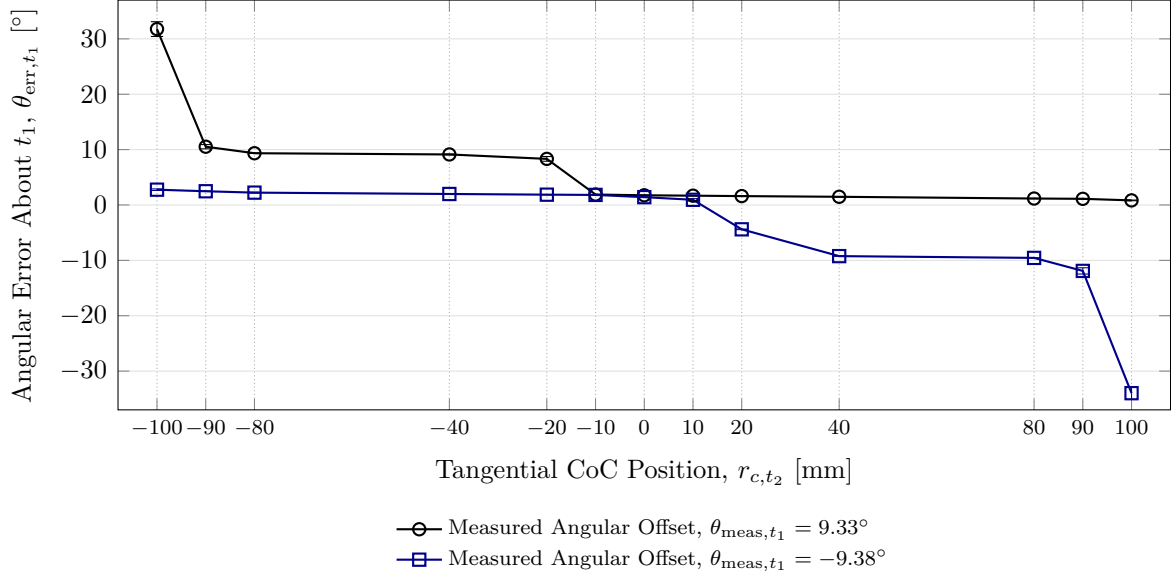


Figure 5.7: Angular error by tangential CoC position.

The angular error depended on the CoC position and entry direction. For the positive-offset condition, $\theta_{\text{meas},t_1} = 9.33^\circ$, Figure 5.7 shows the angular error decreasing from 31.77° at $r_{c,t_2} = -100$ mm to 0.83° at 100 mm. The latter was the smallest mean error magnitude among the tested positions for this entry direction. It was 52.6 % below the 1.75° TCP-centred reference.

For the negative-offset condition, $\theta_{\text{meas},t_1} = -9.38^\circ$, the angular error changed from 2.76° at -100 mm to -33.98° at 100 mm. The smallest measured mean magnitude was 0.94° at 10 mm, which was 33.7 % below the 1.41° TCP-centred reference. At -100 mm, the first-tangent error component crossed zero and ended with a larger magnitude than at the TCP. An increased contact-induced rotation therefore did not necessarily reduce the angular error.

The two entry directions gave their smallest measured mean error magnitudes at different tested positions. A CoC setting that reduces the error for one entry direction can leave a larger error for the other. The largest sample standard deviation was 1.33° , for the positive-offset condition at $r_{c,t_2} = -100$ mm.

To relate the angular error to the interaction wrench, the model-estimated external wrench was evaluated for the positive-offset condition. The three CoC positions were tested with a configured pre-contact offset of $\theta_{\text{offset},t_1} = 10^\circ$. The compared positions were $r_{c,t_2} = -40$, 0 and 40 mm. Each trace shows one measured repetition for the corresponding centre position.

The model-estimated external force and moment supplied by libfranka are evaluated as the measured interaction wrench available from the robot. The est index is retained because they are obtained from the robot's model-based external-wrench estimator rather than from a separate force/torque sensor. The estimated moment is expressed about the TCP before being resolved about the surface tangent, so the three CoC positions are compared at one reference point. The estimated normal-force component and the estimated TCP moment about the first surface tangent are

$$F_{n,\text{est}} = n_s^\top f_{\text{est}}, \quad M_{t_1,\text{est}} = t_1^\top m_{\text{est},\text{TCP}}.$$

The normal-force component is negative while the tool presses into the surface.

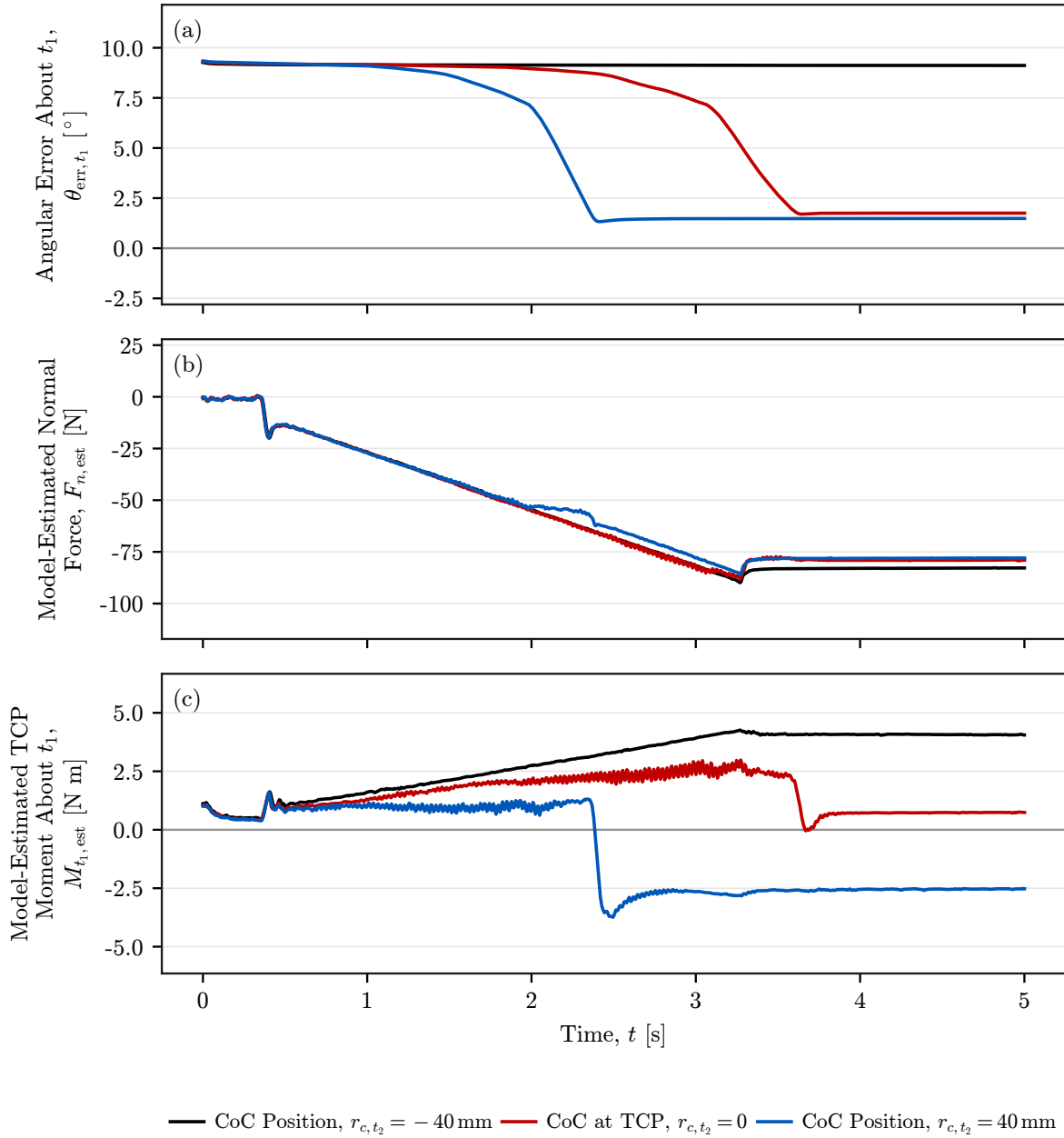


Figure 5.8: Angular error and model-estimated interaction wrench.

Figure 5.8 compares the angular error $\theta_{\text{err}, t_1}(t)$ in panel (a), the model-estimated normal force $F_{n, \text{est}}$ in panel (b), and the model-estimated TCP moment about the first surface tangent $M_{t_1, \text{est}}$ in panel (c), on a common time axis for the three centre positions. Over the final second of contact establishment, the estimated normal force settled at approximately -82.9 N at -40 mm, -79.0 N at the TCP, and -78.0 N at 40 mm. The estimated TCP moment about t_1 settled at approximately 4.07 N m, 0.73 N m and -2.54 N m at the same three positions.

The estimated normal force remained similar for the three CoC positions. The estimated

TCP moment changed substantially and reversed direction between the two displaced positions. The angular errors in the illustrated trials were approximately 9.12° , 1.75° and 1.49° , at -40 mm, the TCP and 40 mm, respectively. The comparison links the changed angular error mainly to the changed contact moment, while the normal force remained similar.

For the two traces in panel (a) of Figure 5.8, the approach to the angular error was compared using a common band. The comparison time is the first recorded instant after which the component stays within 0.1° of its own final value through contact end. This condition was reached at about 2.5 s for $r_{c,t_2} = 40$ mm and 3.6 s at the TCP. The displaced-centre trace reached the band approximately 1.1 s earlier.

At the TCP, $r_c = 0$, so the additional CoC-induced moment vanishes. The rotational spring and damper remain active. A displaced CoC adds a preferred moment direction and changes the angular evolution during contact. Its angular error depends on both the entry direction and how far the measured tool-normal component moves relative to the calibrated reference.

5.3 Null-Space Pose-Hold Results

The redundant joint-space response was measured with a fixed Cartesian pose reference under the same virtual disturbance. Four settings were compared: no null-space torque, projected damping, singular-value conditioning, and combined damping and conditioning. The enabled values were $d_{\text{null}} = 2$ N m s/rad and $k_\sigma = 2$ N m. The commanded disturbance reached 20.00 N in all twelve trials. Peak equivalent joint-torque norms ranged from 2.16 to 2.45 N m, with a torque-limit scale of one throughout the disturbance.

The figures show three-trial means and one sample standard deviation. The measured interval from 5 to 9 s is displayed from 0 to 4 s, relative to disturbance onset. In all twelve trials, the maximum measured TCP position error remained below 2 mm. This criterion assesses position only.

5.3.1 Cumulative Joint Motion

Figure 5.9 shows the cumulative joint motion $E_N(t)$ during the disturbance. Its mean endpoint decreased from 7.60° without null-space torque to 5.69° with projected damping, a reduction of 25.1% relative to the inactive setting. At $k_\sigma = 2 \text{ N m}$, conditioning alone produced 1.69° of cumulative joint motion. Adding projected damping reduced this value to 0.83° , a reduction of 50.7% relative to conditioning alone.

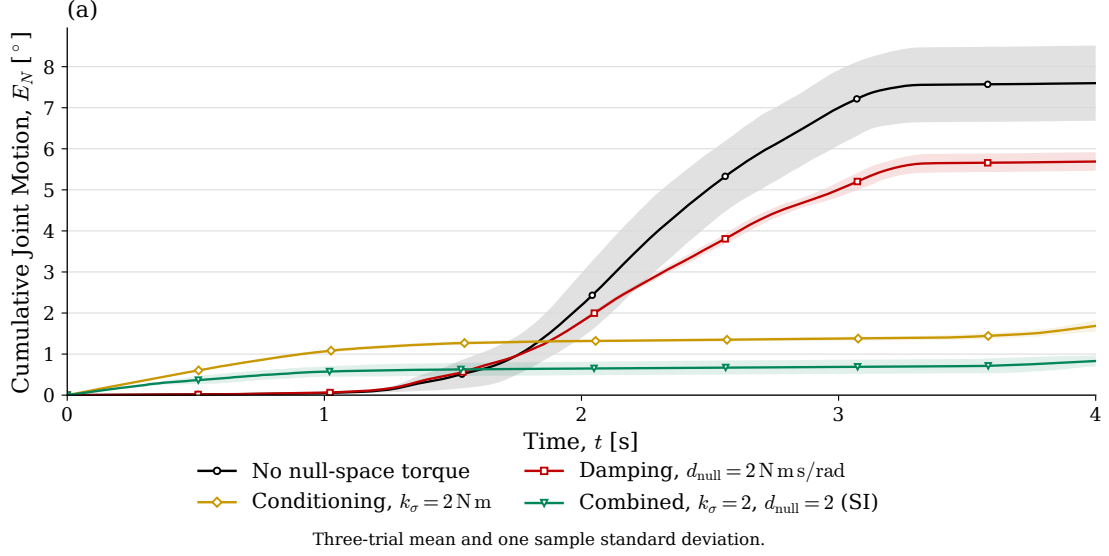


Figure 5.9: Cumulative joint motion during the virtual disturbance.

Projected damping opposes the projected joint velocity. The conditioning term selects a torque direction from the sampled singular-value difference. Both terms act through the null-space torque projector. Their combination reduced the accumulated redundant motion in the tested pose-hold configuration.

5.3.2 Jacobian Conditioning

Figure 5.10 shows the minimum singular value $\sigma_{\min}(t)$ for the same four settings. It decreased during the disturbance without null-space torque and with projected damping alone. With conditioning alone and with combined control, the curves remained close to their respective disturbance-entry values.

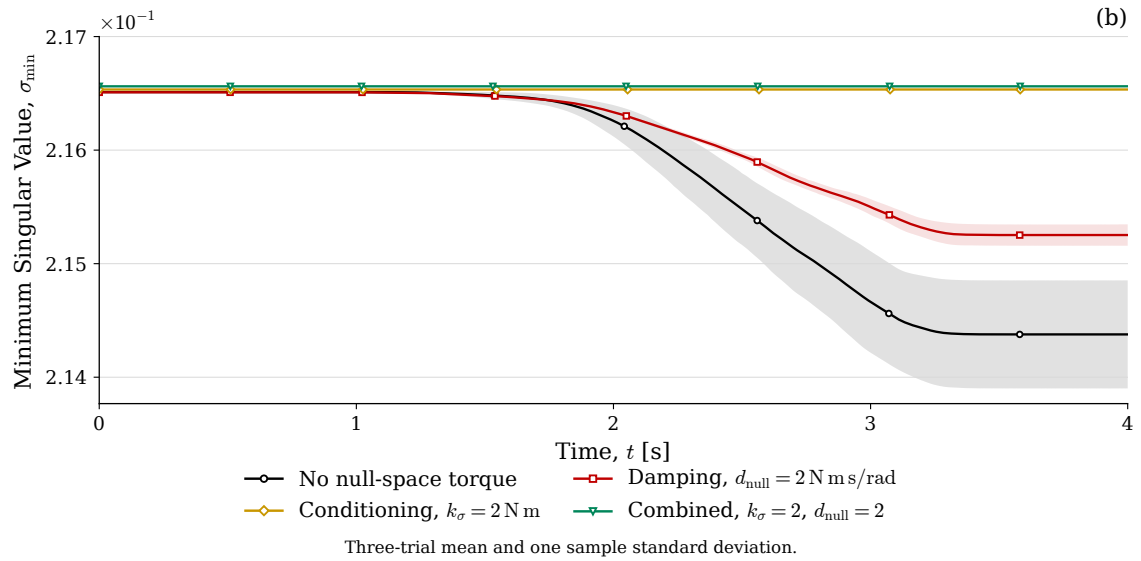


Figure 5.10: Jacobian conditioning during the virtual disturbance.

Conditioning was active during the 5 s preceding the disturbance and remained active throughout it. The comparison therefore includes the redundant configurations reached during this settling interval. The conditioning objective drives the configuration towards a local region with a larger minimum singular value. Adding damping reduced cumulative joint motion while retaining this behaviour in the measured trajectories.

5.3.3 Joint Motion Over Time

Figure 5.11 shows the mean measured joint-1 angle change $\Delta q_1(t)$ relative to disturbance onset. Panel (a) compares all four settings over the complete disturbance interval. Panel (b) enlarges the conditioning and combined-control traces over the same 0–4 s interval. The mean joint-angle change was substantially smaller with either conditioning setting than without conditioning.

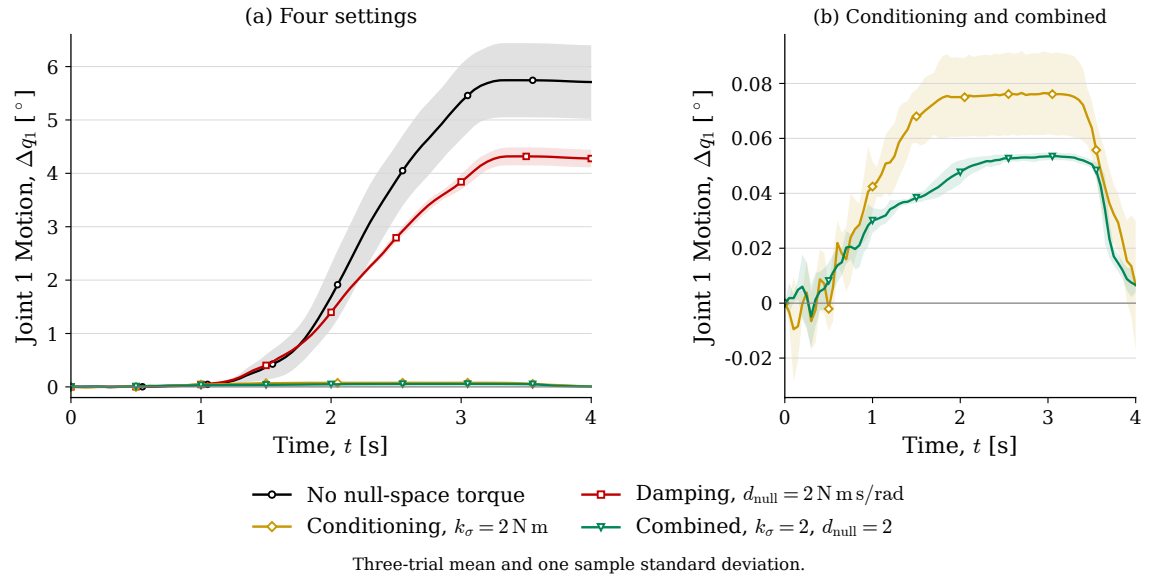


Figure 5.11: Joint motion during the virtual disturbance.

The combined setting showed a smaller mean joint-angle change during the constant-force part of the disturbance than conditioning alone. Both means decreased towards their onset values as the disturbance was released. The shaded bands describe the spread between trials at each instant. The joint-1 trace illustrates the temporal response, while the cumulative projected joint motion in Figure 5.9 evaluates motion across all seven joints.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

A real-time Cartesian impedance controller was implemented on a seven-degree-of-freedom Franka Emika Panda to generate compliant contact-induced rotation towards the configured surface. The controller assigns directional translational and rotational stiffness and damping in the surface frame and allows the virtual centre of compliance to be displaced from the TCP.

Shifting the CoC transforms the Cartesian stiffness and damping matrices and introduces off-diagonal coupling terms between translation and rotation. A translational position error can therefore contribute to the commanded moment, while a rotational error can contribute to the commanded force. The shifted reference also gives the commanded press the virtual lever arm r_c , which generates the additional moment $r_c \times f$ at the TCP.

In the contact measurements, changing the CoC position had little effect on the model-estimated normal force but changed the estimated TCP moment substantially. The angular error $\theta_{\text{err},t_1}(t_{\text{end}})$ changed with this moment. The comparison indicates that the CoC mainly influenced the angular behaviour through its additional commanded moment at the TCP.

With the CoC at the TCP, the angular-error magnitude was smaller than the entry magnitude by 81.2% and 85.0% for the positive- and negative-offset conditions. The negative-offset component crossed zero. The nominal-zero condition instead showed an increase in angular error. At $r_c = 0$, the additional CoC-induced moment vanishes while

the rotational spring and damper remain active. The TCP provides a neutral impedance reference with respect to this added moment.

Higher rotational stiffness left a larger angular error. Increasing K_{R,t_1} from 5 to 50 N m/rad increased the mean angular error by 276.2 % relative to the lowest stiffness. The tested variation of translational stiffness along t_2 produced a span of 3.8 % of its TCP-centred reference error. The translational stiffness along t_1 remained constant. Within these tested ranges, the angular error was more sensitive to rotational stiffness.

The effect of CoC displacement depended on the entry direction. The smallest measured mean error magnitudes were approximately 53 % and 34 % below their respective TCP-centred references, at different tested positions. The first-tangent error component could cross zero. Further contact-induced rotation could increase its final magnitude. The illustrated positive-offset trace reached a band around its angular error earlier with a selected displacement than at the TCP. These results motivate selection based on the measured angular error and its evolution during contact. Returning the CoC to the TCP after alignment would remove the directional preference of the added moment.

The secondary null-space functions were evaluated individually and together with a fixed Cartesian pose reference. Projected damping reduced cumulative joint motion by 25.1 % relative to the inactive setting. At $k_\sigma = 2$ N m, adding damping to conditioning reduced cumulative joint motion by 50.7 % relative to conditioning alone. Both conditioning settings retained the minimum singular value close to its disturbance-entry value. The joint-angle histories showed smaller motion with conditioning than without it under the tested disturbance.

6.2 Limitations

6.2.1 Scope of the Surface-Contact Evaluation

The surface-contact study used one Panda configuration, one rectangular tool and mounting arrangement, one configured surface reference, and one Contact Establishment trajectory. The measured effects therefore apply to this experimental geometry and to the investigated ranges of rotational stiffness, translational stiffness, angular offset, and CoC displacement.

The main repeatable comparison concerned rotation about t_1 . Measurements about t_2 showed greater variability and did not provide the same repeatable response. Each contact trial used one selected CoC displacement throughout Contact Establishment; an adaptive change of the CoC during alignment was not evaluated. The quantitative evaluation focused on Contact Establishment and did not include the subsequent grinding motion.

6.2.2 Calibration and Measurement Scope

The measured angular offset θ_{meas,t_1} and angular error $\theta_{\text{err},t_1}(t)$ use the calibrated surface reference and tool normal with the measured end-effector orientation. The surface reference was established with the tool face seated on the physical plate and provides an estimate of its normal. Its construction used the nominal end-effector axis, while the separate tool-normal calibration found a direction differing by approximately 1.56° . The configured reference retained its original construction.

The instantaneous physical tool–surface angle was not measured independently during contact. The calculated angular error therefore estimates physical misalignment subject to the surface-reference and tool-normal calibration. No numerical uncertainty bound for that estimate was established. The reported component about t_1 also leaves the error about the other tangent outside the principal comparison. A zero first-tangent component alone does not establish complete normal alignment.

The comparison endpoints were stored to 0.01° . A sample standard deviation of zero at this precision indicates unresolved variation in the stored endpoint values. The physical pressure centre and instantaneous contact location were not measured, so their individual contributions to the interaction remain unresolved.

6.2.3 Tool-Mount Compliance

The tool mount exhibited approximately $\pm 2^\circ$ of unloaded angular play about y_{EE} , which is approximately aligned with t_2 in the flat configuration. This mechanical play may have contributed to the greater variability of the t_2 measurements.

The angular error $\theta_{\text{err},t_1}(t)$ assumes the calibrated tool normal remains fixed relative to the end effector. The instantaneous relative tool–gripper rotation was not tracked sepa-

rately during contact. Motion within the mount can therefore make the physical tool-face direction differ from the direction calculated using that calibration.

6.2.4 Scope of the Null-Space Evaluation

Projected damping and singular-value conditioning were evaluated individually and together in Cartesian pose hold under one virtual disturbance direction and one initial robot configuration. The combined-control trials were acquired in a later session with matching saved settings and controller revision. Session effects were not independently controlled. The measured comparisons therefore remain specific to the investigated condition.

6.3 Future Work

For future work, a more rigid tool mounting arrangement would provide a more stable basis for the evaluation about t_2 , where greater variability was observed. The CoC results further motivate an adaptive strategy in which the controller begins with the CoC at the TCP, determines the angular-offset direction, and temporarily applies a tangential CoC displacement that supports the required rotation. After alignment, the CoC can return to the TCP to remove the directional preference and retain compliant behaviour for both rotational directions.

Bibliography

- [1] Alin Albu-Schäffer, Christian Ott, Udo Frese, and Gerd Hirzinger. Cartesian impedance control of redundant robots: Recent results with the DLR-light-weight-arms. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, volume 3, pages 3704–3709, 2003. doi: 10.1109/ROBOT.2003.1242165.
- [2] Benjamin Alt, Florian Stöckl, Silvan Müller, Christopher Braun, Julian Raible, Saad Alhasan, Oliver Rettig, Lukas Ringle, Darko Katic, Rainer Jäkel, Michael Beetz, Marcus Strand, and Marco F. Huber. RoboGrind: Intuitive and interactive surface treatment with industrial robots. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 2140–2146, 2024. doi: 10.1109/ICRA57147.2024.10611143.
- [3] John J. Craig. *Introduction to Robotics: Mechanics and Control*. Pearson Prentice Hall, Upper Saddle River, NJ, 3 edition, 2005.
- [4] Jacques Denavit and Richard S. Hartenberg. A kinematic notation for lower-pair mechanisms based on matrices. *Journal of Applied Mechanics*, 22(2):215–221, 1955.
- [5] Ernest D. Fasse and Jan F. Broenink. A spatial impedance controller for robotic manipulation. *IEEE Transactions on Robotics and Automation*, 13(4):546–556, 1997. doi: 10.1109/70.611320.
- [6] Franka Emika GmbH. libfranka: Communication Test Example. https://github.com/frankarobotics/libfranka/blob/0.7.1/examples/communication_test.cpp, 2019. Official libfranka 0.7.1 example, file `examples/communication_test.cpp`.
- [7] Franka Emika GmbH. Franka control interface documentation. <https://frankaemika.github.io/docs/>, 2023. Accessed for robot interface, libfranka model access, and torque control documentation.

- [8] Franka Emika GmbH. libfranka: Cartesian Impedance Control Example. https://github.com/frankaemika/libfranka/blob/master/examples/cartesian_impedance_control.cpp, 2023. Official libfranka example, file `examples/cartesian_impedance_control.cpp`.
- [9] Christian Friedrich, Patrick Frank, Marco Santin, and Matthias Haag. Highly dynamic physical interaction for robotics: Design and control of an active remote center of compliance. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025. doi: 10.1109/ICRA55743.2025.11128451.
- [10] Richard Hartisch and Kevin Haninger. Flexure-based environmental compliance for high-speed robotic contact tasks. In *2022 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)*, pages 1608–1613, 2022. doi: 10.1109/AIM52237.2022.9863334.
- [11] Neville Hogan. Impedance control: An approach to manipulation—parts i–iii. *Journal of Dynamic Systems, Measurement, and Control*, 107(1):1–24, 1985.
- [12] Oussama Khatib. A unified approach for motion and force control of robot manipulators. *IEEE Journal on Robotics and Automation*, 3(1):43–53, 1987.
- [13] Kevin M. Lynch and Frank C. Park. *Modern Robotics: Mechanics, Planning, and Control*. Cambridge University Press, 2017. URL <https://hades.mech.northwestern.edu/images/7/7f/MR.pdf>. Free preprint.
- [14] Matthias Mayr and Julian M. Salt-Ducaju. A C++ implementation of a cartesian impedance controller for robotic manipulators. *Journal of Open Source Software*, 9(93):5194, 2024. doi: 10.21105/joss.05194.
- [15] Kang Min, Fenglei Ni, Zhaoyang Chen, and Hong Liu. A force control method integrating human skills for complex surface finishing. *Machines*, 12(11):756, 2024. doi: 10.3390/machines12110756.
- [16] Richard M. Murray, Zexiang Li, and S. Shankar Sastry. *A Mathematical Introduction to Robotic Manipulation*. CRC Press, 1994.
- [17] James L. Nevins and Daniel E. Whitney. Computer-controlled assembly. *Scientific American*, 1989. Background source on robotic assembly and compliance.

- [18] Christian Ott, Alin Albu-Schäffer, Andreas Kugi, and Gerd Hirzinger. On the passivity-based impedance control of flexible joint robots. *IEEE Transactions on Robotics*, 24(2):416–429, 2008.
- [19] Bruno Siciliano, Lorenzo Sciavicco, Luigi Villani, and Giuseppe Oriolo. *Robotics: Modelling, Planning and Control*. Springer, London, 2009.
- [20] Stefano Stramigioli, Claudio Melchiorri, and Stefano Andreotti. Intrinsically passive grasping and manipulation. *IEEE Transactions on Robotics and Automation*, 2000. URL <http://robby.caltech.edu/~jwb/courses/CDS270/papers/passive.grasp.pdf>. Open-access preprint.
- [21] Markku Suomalainen, Sylvain Calinon, Emmanuel Pignat, and Ville Kyrki. Improving dual-arm assembly by master-slave compliance. In *2019 International Conference on Robotics and Automation (ICRA)*, pages 8676–8682, 2019. doi: 10.1109/ICRA.2019.8793977.
- [22] Sen Wang, Guodong Chen, Hui Xu, and Zheng Wang. A robotic peg-in-hole assembly strategy based on variable compliance center. *IEEE Access*, 2019. doi: 10.1109/ACCESS.2019.2954459.
- [23] Paul C. Watson. Remote center compliance system. U.S. Patent 4,098,001, 1978. Assigned to The Charles Stark Draper Laboratory.
- [24] Chong Wu, Kai Guo, and Jie Sun. Dual PID adaptive variable impedance constant force control for grinding robot. *Applied Sciences*, 13(21):11635, 2023. doi: 10.3390/app132111635.

Appendix A

Controller Source Listings

Three abridged source listings show the parts of the controller that are specific to this thesis: the compliance-centre point-shift transformation, the selection between the TCP-centred and displaced-CoC impedance, and the final joint-torque assembly. They use the Eigen aliases `Vec3`, `Vec6`, `Vec7`, `Mat3`, `Mat6x6`, and `Mat6x7`. The identifier `R_base_surface` denotes the orientation of the surface frame relative to the robot base frame.

The callback listings follow the official libfranka Cartesian impedance example [8]. The compliance-centre transformation, null-space torque, and experimental disturbance are controller-specific extensions.

A.1 Centre of Compliance and the Offset Adjoint

The thesis defines the compliance-centre displacement as $r_c = p_c - p_{\text{TCP}}$, which points from the TCP to the selected centre of compliance. The controller source uses the same vector, so the code identifier `r_c` in Listings A.1 and A.2 is the thesis quantity. Its upper-right block $-[r_c]_\times$ is the upper-right block of the adjoint defined in Equation (2.37). The same point-shift transformation is applied to the Cartesian stiffness and damping matrices.

Listing A.1: Source-side compliance-centre adjoint and point-shift transform.

```
1 | Mat3 skewMatrix(const Vec3& v) {  
2 |     Mat3 s;  
3 |     s << 0.0, -v(2),  v(1),  
4 |         v(2),  0.0, -v(0),  
5 |         -v(1),  v(0),  0.0;  
6 |     return s;  
7 | }
```

```

7 | }
8 |
9 | Mat6x6 complianceShiftMatrix(const Vec3& r_c) {
10 |     // Defining the spatial shift for r_c = p_C - p_TCP [m].
11 |     Mat6x6 adjoint = Mat6x6::Zero();
12 |     adjoint.block<3, 3>(0, 0) = Mat3::Identity();
13 |     adjoint.block<3, 3>(0, 3) = -skewMatrix(r_c);
14 |     adjoint.block<3, 3>(3, 3) = Mat3::Identity();
15 |     return adjoint;
16 | }
17 |
18 | Mat6x6 shiftGainToTcp(
19 |     const Mat6x6& gain_at_center, const Vec3& r_c) {
20 |     const Mat6x6 adjoint = complianceShiftMatrix(r_c);
21 |     return adjoint.transpose() * gain_at_center * adjoint;
22 | }
    
```

The tool-frame parameter `compliance_center_offset_ee` stores the displacement r_c in end-effector coordinates, and the controller rotates it into the base frame without changing its sense. The alternative surface-frame parameter `compliance_lever_surface` stores the same displacement in (t_1, t_2, n_s) components. All non-zero CoC displacements used in the experiments were configured through the tool-frame definition.

Listing A.2 shows how the controller selects between the block-diagonal impedance at the TCP and the point-shifted impedance about a displaced centre of compliance.

Listing A.2: Selection of the decoupled or coupled spring wrench (abridged).

```

1 | const bool coupled =
2 |     params.use_virtual_compliance_center &&
3 |     (state == ControlState::kContactEstablishment ||
4 |     state == ControlState::kPreGrindingHold ||
5 |     (state == ControlState::kPoseHold &&
6 |     params.use_contact_impedance_hold));
7 | if (!coupled) {
8 |     // Two independent springs. dv.tail is -omega, so the damping
9 |     // signs match.
10 |     Vec6 wrench;
11 |     wrench.head<3>() = Kp * dx.head<3>() + Dp * dv.head<3>();
12 |     wrench.tail<3>() = KR * dx.tail<3>() + DR * dv.tail<3>();
13 |     return wrench;
14 | }
15 |
16 | // Resolving r_c = p_C - p_TCP in the robot base frame [m].
17 | Vec3 r_c = Vec3::Zero();
18 | if (params.compliance_center_in_tool_frame) {
19 |     r_c = R_EE * params.compliance_center_offset_ee;
20 | } else {
21 |     r_c = R_base_surface * params.compliance_lever_surface;
22 | }
23 | const Mat6x6 K_tcp =
24 |     shiftGainToTcp(blockDiagonal(Kp, KR), r_c);
25 | const Mat6x6 D_tcp =
26 |     shiftGainToTcp(blockDiagonal(Dp, DR), r_c);
27 | return K_tcp * dx + D_tcp * dv;
    
```

A.2 Final Torque Assembly

Listing A.3 shows how the callback combines the Cartesian impedance wrench, the null-space torque, the experiment-specific disturbance, and the model Coriolis contribution

into the commanded joint torque of Equation (2.73). The acquisition of the measured joint state, the Jacobian and the end-effector pose is standard libfranka usage and is described in Section 3.1.

Listing A.3: Final joint-torque assembly (abridged).

```

1 | const Vec6 wrench =
2 |     computeCartesianImpedanceWrench(
3 |         params, state, Kp_used, Dp_used,
4 |         KR_used, DR_used, R_base_surface,
5 |         dx, dv, R_EE);
6 |
7 | const Vec7 tau_nullspace = computeNullspaceTorque(
8 |     params, model, state, J, dq);
9 |
10 | const Vec7 tau_task = J.transpose() * wrench;
11 | const Vec7 tau_cmd =
12 |     tau_task + tau_nullspace + disturbance.tau + coriolis;

```

The Cartesian pose-hold study of Section 4.7 activates the disturbance term. In the surface-contact experiments this contribution is zero.

Appendix B

Recorded Experimental Data

During torque control, the controller stores the experimental signals in a preallocated ring buffer inside the 1 kHz callback. After the controller stops, the retained rows are written to a CSV file in chronological order.

B.1 Evaluation Signals

Table B.1 lists the recorded signal groups used to calculate the reported surface-contact and null-space results. The surface-contact schema supplies the angular-error, contact-geometry, commanded-wrench and model-estimated wrench groups, and the Cartesian pose-hold schema supplies the singular-value, projected-motion, joint-angle, and disturbance groups. The complete file schemas, including the diagnostic fields that no reported result uses, are documented with the recorded data.

A suffix `_x`, `_y`, or `_z` denotes a base-frame Cartesian component, and a suffix `_1` to `_7` denotes a joint component.

Table B.1: Recorded signal groups used in the evaluation.

Column group	Meaning
<code>time, state</code>	Elapsed time and active control state.
<code>p_EE, p_d</code>	Measured and desired TCP positions [m].

Table B.1 continued

Column group	Meaning
<code>tool_contact,</code> <code>edge_target</code>	Measured position of the selected geometric tool point and its desired contact-establishment target, in the base frame [m].
<code>tool_contact_offset_ee</code>	Offset of the selected geometric tool point from the TCP, in end-effector coordinates [m].
<code>e_p,</code> <code>e_R,</code> <code>pdot,</code> <code>pdot_d,</code> <code>omega</code>	Cartesian position and orientation errors, measured TCP velocity, desired TCP velocity, and measured end-effector angular velocity. During Contact Establishment, <code>e_R</code> is calculated relative to the orientation captured at the clearance transition.
<code>angular_deviation_</code> <code>t1_deg</code>	Normal-to-normal angular-error component about t_1 , calculated from the measured end-effector orientation and calibrated tool normal $[\circ]$. Its recorded direction is from the tool normal towards $-n_s$.
<code>p_CoC</code>	Base-frame CoC position $p_c = p_{\text{TCP}} + r_c$ [m], recorded for every surface-contact trial, including the TCP-centred condition $r_c = 0$.
<code>f,</code> <code>m</code>	Complete commanded Cartesian force and TCP moment, in the base frame [N], [Nm].
<code>external_force,</code> <code>external_moment</code>	Model-estimated external wrench supplied by libfranka. It acts on the stiffness frame $\{K\}$ and is expressed in the base frame, and the recorded moment is referenced to the base origin [N], [Nm].
<code>nullspace_mode</code>	Selected null-space law.
<code>sigma_current</code>	Current smallest singular value $\sigma_{\min}$; present in the Cartesian pose-hold schema.

Table B.1 continued

Column group	Meaning
sigma_plus, sigma_minus, sigma_difference, sigma_direction	Smallest singular values at the two sampled configurations $q \pm \alpha_{\text{probe}} v_7$, their difference, and the resulting direction selector. These fields occur only in the Cartesian pose-hold schema.
phase_time_s, q1_deg	Elapsed pose-hold time [s] and measured joint 1 angle q_1 [°], from the separate joint-state diagnostic file sampled at a nominal 20 Hz.
nullspace_dq_1..7	Components of the projected joint velocity $\dot{q}_{\text{null}}$ [rad/s].
tau_sigma_1..7, tau_sigma_norm, tau_nullspace_norm	Singular-value torque τ_σ and total null-space torque τ_{null} [N m].
nullspace_damping	Active projected damping coefficient [N m s/rad].
disturbance_scale, disturbance_torque_scale, disturbance_force_ base_x..z, disturbance_ point_base_x..z	Commanded virtual-disturbance waveform s_{dist} , applied torque-limit scale, commanded base-frame point force [N], and the base-frame position of the selected point on link 3 [m].
tau_disturbance_1..7	Components of the equivalent joint disturbance torque $\tau_{\text{dist}} = J^\top f_{\text{dist}}$ [N m].

The angular-error evaluation of Section 4.5.2 takes the negative of `angular_deviation_t1_deg` to obtain θ_{err,t_1} in degrees. Earlier contact files name the same quantity `alignment_error_t1_deg`. The recorded component is evaluated directly at each instant. For the comparison tables, the terminal controller reports supply the contact-entry and contact-end components to 0.01°. Each report uses the same normal-to-normal calculation as the time-series field.

The Cartesian wrench evaluation of Section 4.6 reads the model-estimated group. Because the recorded moment is referenced to the base origin, the evaluation expresses it about the TCP, $m_{\text{est},\text{TCP}} = m_{\text{est}} - p_{\text{TCP}} \times f_{\text{est}}$, before resolving $M_{t_1,\text{est}}$ about the first surface

tangent. In the experimental configuration ${}^{\text{EE}}T_K = I$, so the stiffness frame coincides with the TCP and no further frame change is required.

Appendix C

Controller Parameters

This appendix collects the numerical controller, geometry, state-transition, null-space, and operating parameters used in the experiments described in Chapters 3 and 4.

C.1 Surface and Tool Geometry

The configured surface reference satisfies

$$n_s^\top(p - p_s) = 0, \quad n_s = R_y(\beta_s)R_x(\alpha_s)e_{z_0},$$

and the surface-frame axes are ordered as $[t_1, t_2, n_s]$. Table C.1 lists the corresponding surface, tool-normal, and rectangular tool-geometry parameters.

Table C.1: Surface and tool parameters.

Quantity	Configuration key	Value
Surface point	<code>surface_point</code>	$[0.515267, -0.107172, 0.003108]$ m
Surface tilts	α_s, β_s	$-1.585^\circ, +0.988^\circ$
In-plane reference direction	<code>surface_tangent1_hint_base</code>	$[1, 0, 0]$
Tool axis in EE	<code>tool_axis_ee</code>	$[0.026387057, -0.006678514, 0.999629492]$
Tool-normal target direction	<code>tool_axis_target_sign</code>	$-n_s$
Tool-axis spin angle measured from t_1	<code>tool_target_offset_normal_deg</code>	90.0°
Tool-face centre in EE	<code>tool_contact_face_center_ee</code>	$[0, 0, 0.020]$ m
Tool face size	width $\times$ length	0.040×0.120 m
Tool-point selection tolerance, ε_{sel}	<code>tie_tolerance</code>	0.1 mm

C.1.1 Calibration Relations

For the seated surface-reference pose, the nominal end-effector Z_{EE} axis and the calibrated tool-face-centre offset were used to calculate the stored plane quantities. The tool face rests against the plane with $+Z_{\text{EE}}$ directed into it, so the outward normal is the opposite direction,

$$n_s = -R_{\text{EE}}[0, 0, 1]^\top, \quad p_s = p_{\text{EE}} + R_{\text{EE}}r_{\text{face,EE}}. \quad (\text{C.1})$$

The first tangent was obtained by projecting the base-frame x_0 direction onto the configured plane and normalising the result, and $t_2 = n_s \times t_1$ completed the surface frame. For the tool-normal calibration, the fitted direction was

$$n_{\text{Tool,EE}} = \arg \min_{\|\tilde{n}\|=1} \sum_{i < j} \|(R_{\text{cal},i} - R_{\text{cal},j})\tilde{n}\|^2. \quad (\text{C.2})$$

Three seated yaw captures entered the fit. The sign of the fitted eigenvector was selected towards nominal $+Z_{EE}$.

C.1.2 Initial Joint Configuration

The initial joint configuration used in the experiments is

$$q_{\text{init}} = [0.014777, 0.210911, -0.009320, -2.284835, \\ -0.002725, 2.493410, 0.780694]^\top \text{ rad.}$$

C.2 State Sequence and Impedance

The contact experiments used inertia-scaled damping during Surface Approach and Contact Establishment. The damping vectors listed for those two states in Table C.2 are therefore the configured fallback values, applied when the calculated coefficient fails its validity check. The Cartesian pose-hold trials used the fixed damping matrices of the final rows.

Table C.2: State-specific stiffness and fallback damping parameters.

State / quantity	Active stiffness and frame	Damping role and value
Approach translation	$[150, 150, 150] \text{ N/m}$, surface $[t_1, t_2, n_s]$	Fallback: $[20, 20, 20] \text{ N s/m}$
Approach rotation	$[180, 180, 90] \text{ N m/rad}$, surface $[t_1, t_2, n_s]$	Fallback: $[15, 15, 12] \text{ N m s/rad}$
Contact Establishment translation	$[2000, 2000, 350] \text{ N/m}$, surface $[t_1, t_2, n_s]$	Fallback: $[10, 10, 175] \text{ N s/m}$
Contact Establishment rotation	$[5, 5, 50] \text{ N m/rad}$, surface $[t_1, t_2, n_s]$	Fallback: $[10.01, 10.01, 10] \text{ N m s/rad}$
Hold translation	2000 N/m , isotropic base	Fixed: 50 N s/m
Hold rotation	50 N m/rad , isotropic base	Fixed: 8 N m s/rad

The state-transition and trajectory values applied throughout the surface-contact experiments are given in Table 4.3, and are not repeated here.

C.3 Damping and Centre of Compliance

The thesis defines the compliance-centre displacement as $r_c = p_c - p_{\text{TCP}}$, expressed in the robot base frame. For a tool-fixed centre, the configuration specifies its end-effector components $r_{c,\text{EE}}$, and the controller calculates $r_c = R_{\text{EE}}r_{c,\text{EE}}$. The damping calculation and the compliance-centre transformation are defined in Sections 2.6.1 and 2.7. The tested components are listed in Table 4.4.

Table C.3: Damping and compliance-centre parameters.

Quantity	Symbol / key	Value
Approach damping factor	ζ	1.9
Contact-establishment damping factor	ζ	1.0
Operational-space inertia regularisation, ε	added to $J(q)M(q)^{-1}J^\top(q)$	10^{-9}
Upper bound on each calculated damping coefficient	<code>auto_damping_max</code>	400 N s/m for translation; 400 N m s/rad for rotation
Additional lower bound from manual damping configuration	<code>auto_damping_</code> <code>min_from_manual</code>	0; no additional manual lower bound was applied, and the configured matrices remained available as fallback values
States with active inertia-scaled damping	—	Surface Approach and Contact Establishment
Inertia-scaled damping in pose hold	<code>hold_auto_damping</code>	0 for all twelve pose-hold trials
Point-shifted coupled Cartesian impedance	<code>use_virtual_</code> <code>compliance_center</code>	Enabled during Contact Establishment for the displaced-CoC conditions
Frame in which the configured displacement is fixed	<code>compliance_center_</code> <code>in_tool_frame</code>	Tool-fixed in all surface-contact experiments
Tool-fixed displacement	<code>compliance_center_</code> <code>offset_ee</code>	in end-effector axes

C.4 Null-Space and Operating Parameters

Table C.4 lists the null-space, data-recording, gripper, and robot-side collision parameters used in the experiments. Sections 4.3.2 and 4.7.1 define the surface-contact and pose-hold null-space configurations, respectively.

Table C.4: Null-space and operating parameters.

Parameter	Symbol / key	Value
Null-space mode	<code>nullspace_mode</code>	3 (projected damping and singular-value conditioning)
Null-space damping	d_{null}	2.0 N m s/rad
Sigma torque magnitude	k_{σ}	1.0 N m
Probe distance	α_{probe}	0.08 rad
Singular-value difference deadband	δ_{σ}	10^{-6}
Relative SVD cutoff	ρ	10^{-4}
Gripper grasp	width, speed, force	0.066 m, 0.050 m/s, 60 N
Joint-side collision threshold	acceleration/nominal	140 N m on each axis
Cartesian collision threshold	acceleration/nominal	240 N or N m on each channel

Appendix D

Numerical Results for Main Surface-Contact Cases

Tables D.1 to D.4 give the arithmetic mean and sample standard deviation over the three repeated trials of each setting in Cases A–D. In Table D.4, r_{c,t_2} denotes the tangential CoC coordinate along t_2 , perpendicular to the reported angular-error component about t_1 .

Table D.1: Case-A angular error at the TCP.

Measured Angular Offset	Tangential CoC Position	Angular Error θ_{err,t_1} [°]
$\theta_{\text{meas},t_1} = 0.69^\circ$	TCP	1.65 ± 0.01
$\theta_{\text{meas},t_1} = 9.31^\circ$	TCP	1.75 ± 0.00
$\theta_{\text{meas},t_1} = -9.41^\circ$	TCP	1.41 ± 0.06

Table D.2: Case-B angular error by rotational stiffness.

Measured Angular Offset	CoC Position	K_{R,t_1} [N m/rad]	Reference	Angular Error θ_{err,t_1} [°]
9.31° about t_1	TCP	5	Case-A reference	1.75 ± 0.00
9.32° about t_1	TCP	15	—	2.04 ± 0.02
9.31° about t_1	TCP	50	—	6.58 ± 0.03

Table D.3: Case-C angular error by translational stiffness along t_2 .

Measured Angular Offset	CoC Position	K_{p,t_2} [N/m]	Reference	Angular Error θ_{err,t_1} [°]
9.27° about t_1	TCP	300	—	1.68 ± 0.01
9.24° about t_1	TCP	800	—	1.69 ± 0.01
9.31° about t_1	TCP	2000	Case-A reference	1.75 ± 0.00

Table D.4: Case-D angular error by tangential CoC position.

r_{c,t_2} [mm]	Angular Error θ_{err,t_1} [°]	
	$\theta_{\text{meas},t_1} = 9.33^\circ$	$\theta_{\text{meas},t_1} = -9.38^\circ$
−100	31.77 ± 1.33	2.76 ± 0.14
−90	10.52 ± 0.39	2.48 ± 0.03
−80	9.35 ± 0.02	2.23 ± 0.08
−40	9.13 ± 0.07	1.99 ± 0.02
−20	8.33 ± 0.42	1.86 ± 0.01
−10	1.89 ± 0.01	1.82 ± 0.01
0 (TCP)	1.75 ± 0.00	1.41 ± 0.06
10	1.68 ± 0.01	0.94 ± 0.21
20	1.60 ± 0.01	-4.40 ± 0.06
40	1.48 ± 0.01	-9.24 ± 0.03
80	1.17 ± 0.03	-9.56 ± 0.03
90	1.11 ± 0.06	-11.91 ± 0.57
100	0.83 ± 0.06	-33.98 ± 1.16